

Integrating computational fluid dynamics (CFD) and machine learning to improve urban green infrastructure for heat mitigation and air quality: A systematic review

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ABSTRACT

Urban population growth and rapid urbanization pose significant sustainability challenges, notably intensified urban heat and air pollution. Strategically deployed urban green infrastructure (UGI), such as green roofs, street trees, and urban parks, can effectively mitigate these challenges by reducing air temperature and improving air quality. Deploying UGI optimally requires tools that capture the complex transport of heat and pollutants around vegetation while rapidly exploring large design spaces. Combining high-fidelity computational fluid dynamics (CFD) with data-driven machine learning (ML) offers exactly this capability. Yet, despite advances in digital technologies, the integration of ML and CFD to enhance UGI performance remains relatively underexplored, with existing studies lacking comprehensive analyses of this synergy. This review systematically explores CFD-ML integration in UGI research, demonstrating how it simultaneously addresses computational efficiency and predictive accuracy in urban heat mitigation and air quality improvement. Current CFD-ML integration methods include two main frameworks: direct CFD-ML coupling and ML-based surrogate CFD modeling. Four key application areas for UGI are identified: (1) optimizing UGI design, (2) accelerating simulations, (3) improving understanding of urban microclimate physics, and (4) enhancing CFD simulation quality. The review summarizes frequently used ML algorithms, illustrating significant improvements achieved through CFD-ML integration, such as predictions up to 800 times faster with surrogate CFD-ML approaches while maintaining prediction accuracies of $R^2 \geq 0.90$. Specifically, ML methods have accurately predicted particulate matter concentrations with errors below 10% compared to measured data, and random forest (RF) algorithms trained on ENVI-met simulations achieved cooling load prediction accuracies of up to $R^2 = 0.98$. Additionally, integrated ML frameworks significantly accelerated urban thermal analyses from over 400,000 h to approximately one hour. Challenges remain, including coupling disparate tools, the absence of standardized or generalizable models, and data scarcity for training robust ML models. This study highlights critical research gaps, emphasizes CFD-ML integration benefits, and offers guidance on algorithm selection and integration strategies. Future research directions, including large language model (LLM)-driven CFD, hybrid physics-informed machine learning, transfer learning, and multi-fidelity modeling, are discussed as promising avenues for enhancing CFD-ML integration. Ultimately, utilizing complementary strengths of CFD and ML can produce more efficient, accurate urban climate simulations, offering substantial decision support for UGI planning in mitigating urban heat and improving air quality.

Abbreviations: ANN, artificial neural network; ANFIS, adaptive neuro-fuzzy inference system; AMR, adaptive mesh refinement; CFD, computational fluid dynamics; CNN, convolutional neural networks; CD, cross decomposition; CAE, convolutional autoencoder; DT, decision tree; DRL, deep reinforcement learning; EBM, explainable boosting machine; FNO, Fourier neural operator; FVM, finite volume method; GR, green roofs; GW, green walls; GIS, geographical information systems; GCI, grid convergence index; GPR, Gaussian process regression; GANs, generative adversarial networks; KNN, K-nearest neighbors; LES, large eddy simulation; LAD, leaf area density; LAI, leaf area index; LR, linear regression; LOR, logistic regression; LGBM, light gradient boosting machine; LLMs, large language models; ML, machine learning; MLP, multi-layer perceptron; NIROM, non-intrusive reduced-order modeling; PINN, physics-informed neural network; POD, proper orthogonal decomposition; RANS, Reynolds-Averaged Navier-Stokes; RNN, recurrent neural networks; ROMs, reduced-order models; RL, reinforcement learning; RT, regression tree; RF, random forest; ST, street trees; SR, symbolic regression; SVM, support vector machine; t-SNE, t-distributed stochastic neighbor embedding; UGI, urban green infrastructure; URP, urban forests and parks; UW, urban water; VAEs, variational autoencoders; XGBoost, extreme gradient boosting.

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1. Introduction

As the global population grows and urbanization accelerates, cities face increasingly severe environmental challenges. Currently, urban areas are home to 56% of the world's population, about 4.4 billion people. This number is expected to increase significantly, reaching nearly 7 in 10 people by 2050 [1]. This urban expansion has intensified issues such as urban heat islands and air pollution. Urban heat islands are characterized by significantly higher temperatures in cities compared to rural areas [2], correlating strongly with increased heat-related illnesses and mortality, including heat stress and cardiovascular complications [3]. Across published case studies, this phenomenon has been shown to increase building cooling energy consumption by a median of approximately 19% [4]. On the other hand, air pollution is a major contributor to premature human mortality and has detrimental effects on ecosystems, including reduced biodiversity, soil degradation, and impaired water quality [5].

To address these challenges, urban green infrastructure (UGI) has been increasingly adopted in urban planning. UGI refers to a strategic approach to developing interconnected networks of multifunctional blue and green spaces, utilizing nature-based solutions that integrate vegetation, water, and soil to tackle a wide range of environmental, social, and climate-related challenges [6]. A key strength of UGI lies in its multifunctionality, offering solutions that extend beyond addressing isolated issues. For instance, vegetation improves air quality, provides cooling and shade during hot days, reduces surface water runoff, and enhances biodiversity by creating habitats and food sources for wildlife such as insects and birds [7]. UGI offers integrated benefits across multiple domains: environmental (biodiversity, air pollution mitigation, local climate regulation, soil conservation), social (aesthetic value, opportunities for social interaction, public health), and economic (property value, tourism development) [8]. As illustrated in Fig. 1, UGI types, including street trees (ST), green roofs (GR), green walls (GW), urban forests and parks (URP), and urban water bodies (UW), serve as effective

measures for mitigating urban heat [9], improving air quality [10,11], and enhancing the cultural, aesthetic, and economic value of urban areas. Recent systematic reviews highlight heat mitigation and air quality regulation as two particularly critical ecosystem services that are widely investigated in the context of UGI [12–14].

In response to the critical need for effective urban solutions targeting heat mitigation and air quality improvement, various tools have been adopted to investigate and optimize UGI configurations, including wind tunnel experiments, field measurements, computational fluid dynamics (CFD), and remote sensing techniques [15]. Among these methods, CFD is widely used for simulating airflow, heat transfer, and pollutant dispersion in urban environments, helping analyze air quality and thermal comfort. Previous studies demonstrated UGI's effectiveness in mitigating nighttime urban heat islands [16], reducing NO₂ and particulate matter concentrations [17], and improving the thermal comfort of pedestrians and cyclists [18].

However, CFD faces limitations in complex and large-scale urban simulations. One significant challenge is the substantial computational power required to model turbulence, multiphase interactions, thermal radiation, pollutant dispersion, vegetation effects, solar radiation, and wind dynamics [19]. Accurate simulations of these interactions increase computational costs, especially when fine mesh calculations are necessary [20]. Urban microscale modeling often focuses on capturing general trends rather than precise details, but oversimplification can lead to significant inaccuracies for critical factors like airflow, pollutant dispersion, or heat transfer. For example, Cui et al. [21] used wind tunnel and CFD simulations to study vegetation impacts on contaminant dispersion, but simplifying vegetation into basic geometric shapes reduced the models' accuracy, limiting their reflection of real-world conditions and the practical applicability of the findings.

In addition to heavy computational demands, CFD encounters difficulties in obtaining accurate input data and frequently oversimplifies complex physical processes [22]. These limitations can reduce the precision of simulations, leading researchers to explore strategies to lower

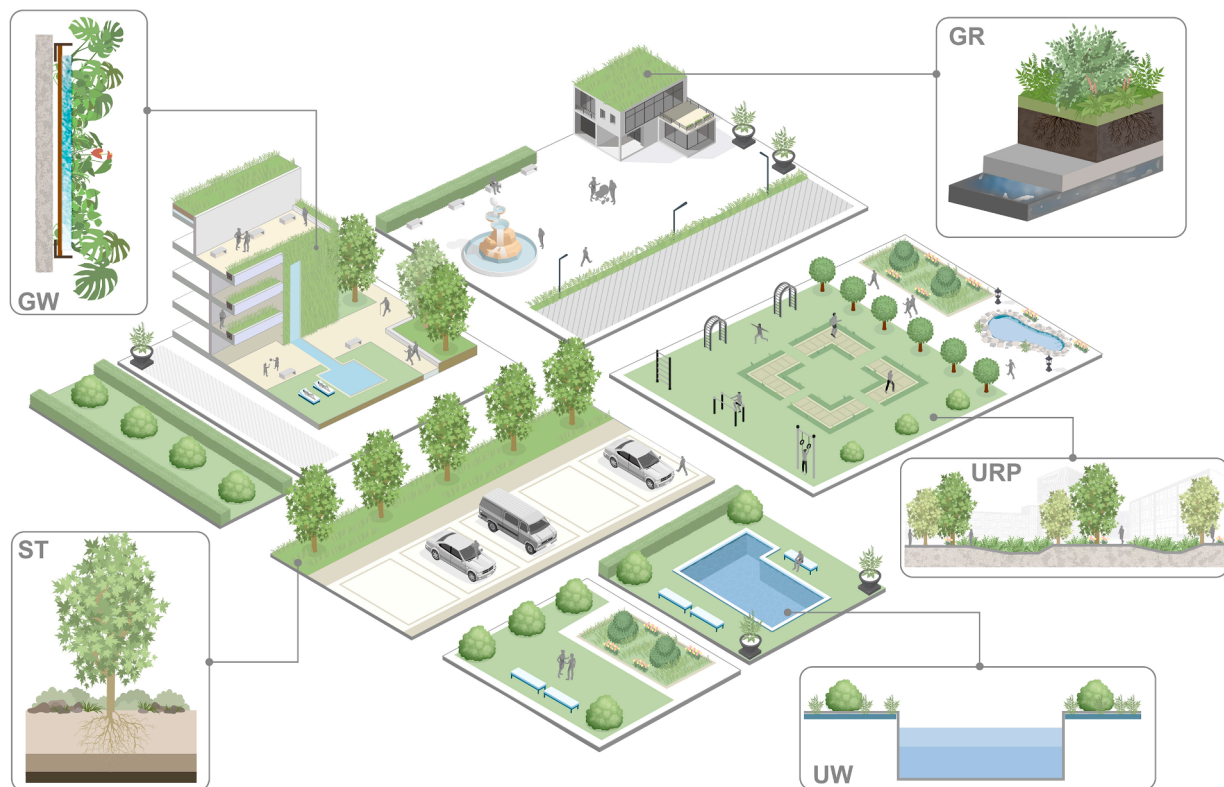


Fig. 1. Five UGI types evaluated in this study: street trees (ST), green roofs (GR), green walls (GW), urban forests and parks (URP), and urban water bodies (UW).

computational costs without sacrificing accuracy [23–25]. To tackle these challenges, many studies have combined CFD with optimization techniques, such as genetic algorithms applied to response-surface or other surrogate models to search the design space efficiently while minimising the number of costly CFD runs [26,27]. Moreover, the growing need for more efficient modeling has increased interest in integrating machine learning (ML) with CFD. This approach aims to improve performance by streamlining simulations and reducing computational load, while maintaining the accuracy needed for reliable predictions in urban environments.

ML technology has rapidly grown due to its ability to learn patterns from large datasets with limited explicit programming. This makes it particularly effective for quickly analyzing and interpreting complex data, providing researchers with novel approaches for both analysis and prediction [28–30]. ML techniques have shown promise in reducing computational demands by using data-driven approaches to approximate some of the complex physical processes traditionally handled by CFD. As a result, researchers are increasingly exploring the use of ML to supplement CFD calculations, with some studies investigating the potential for ML to replace specific aspects of CFD in certain scenarios

[31–33].

For example, predicting the impact of ST on pollutant dispersion might take over 11 h per CFD simulation, whereas an ML model can evaluate multiple scenarios in minutes, offering efficient recommendations [34]. ML models trained on CFD-generated data can quickly predict wind speed, air temperature, and environmental variables, and identify optimal urban structures [35] and vegetation patterns [36]. However, it's worth noting that while ML models offer significant time savings once trained, the training process itself can be computationally demanding, especially when large and complex datasets are required. Additionally, fluid dynamics' complexity and variability mean no single ML model suits all scenarios, leading to hybrid approaches combining traditional mathematical models and ML algorithms [37].

Given these developments, a thorough review of how ML supports CFD simulations in UGI research is crucial. Such a review would highlight the strengths and limitations of various methods, offering insights into integrating CFD domain knowledge with ML algorithms. This would advance urban planning and environmental simulations, enhancing the efficiency of analyzing UGI's role in heat mitigation and air quality improvement. A comprehensive literature review would also help

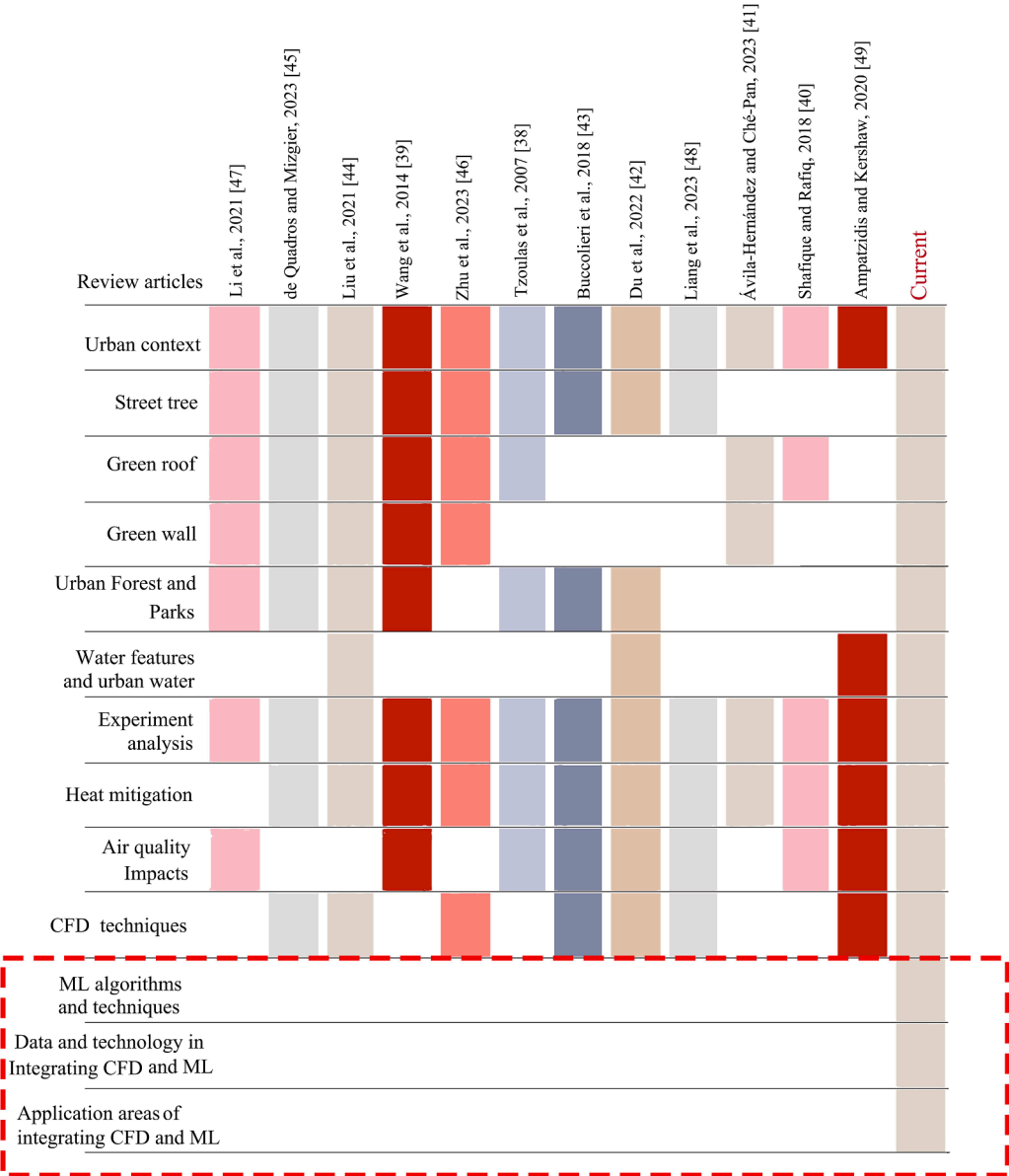


Fig. 2. Coverage of review papers on UGI [38–49].

establish methodologies that effectively utilize these integrated technologies for sustainable urban development.

1.1. Previous related review studies and research gaps

Over the past 17 years, numerous review studies have explored different aspects of UGI, climate change mitigation, and modeling approaches; however, each has maintained a distinct focus, and none have specifically targeted the integration of CFD and ML for urban green infrastructure. Fig. 2 summarizes various research directions explored (and unexplored) by these reviews in a matrix format. Early reviews primarily concentrated on the ecological and human health benefits of UGI [38], as well as its economic value and implications for indoor environments [39]. Subsequent studies expanded the scope, incorporating detailed analyses of specific UGI types, such as comprehensive assessments of the environmental benefits of UGI interventions on urban climate and air quality [40–42]. These reviews employed empirical evidence and traditional modeling approaches to examine how urban greening influences cooling effects or pollutant dispersion at different scales. Other reviews have focused specifically on computational modeling techniques applied in urban environmental contexts [43–47].

For instance, Buccolieri et al. [43] provided an extensive review of vegetation representation (mainly trees) in CFD simulations, exploring its effects on urban air quality and thermal conditions. Their work summarized various CFD parameterization methods for processes such as pollutant deposition, canopy-induced momentum reduction, and vegetation cooling, offering valuable insights into purely CFD-based analyses. Meanwhile, Liang et al. [48] systematically reviewed empirical studies examining how the configuration of urban green spaces, such as the size, shape, and distribution of parks and tree canopies, affects cooling effectiveness. This review synthesized observational and measurement-based findings related to UGI's role in mitigating urban heat islands. While these and other earlier reviews [38–49] provide significant context from vegetation-related CFD modeling to real-world evidence of UGI effectiveness, none have investigated the emerging synergy between CFD and ML methodologies.

In particular, existing reviews have not systematically explored how ML techniques can be effectively integrated with high-fidelity CFD models. There remains a notable research gap regarding detailed analyses of suitable ML algorithms combined with specific CFD models, appropriate application scenarios, data requirements (e.g., type and scale of training datasets), integration strategies, and methods to ensure the reliability and generalizability of simulation results. Addressing these aspects is essential for effectively optimizing UGI designs and accurately evaluating their performance in urban heat mitigation and air quality improvement. Given the growing number of studies integrating ML and CFD to address increasingly complex urban environmental challenges related to heat and air pollution, this gap is particularly critical and warrants systematic synthesis and in-depth exploration.

To address this identified gap, a more comprehensive understanding of ML methodologies and their integration strategies with CFD models is urgently required. Such enhanced integration will support a more holistic and precise assessment of UGI performance. Effective integration of CFD and ML demands careful consideration of model accuracy, appropriate numerical representations of UGI components (such as vegetation and urban water bodies), and compatibility among computational tools, algorithms, and methodological frameworks. Furthermore, there is a need for clearer application examples, standardized guidelines, and best-practice recommendations for effectively combining ML technologies with CFD simulations. Establishing such standards and guidelines will facilitate broader adoption and ensure robustness and reliability in future urban environmental studies.

1.2. Aims and objectives, structure

This study aims to systematically review the integrated application

of CFD and ML techniques for heat mitigation and air quality improvement in UGI research, while addressing the current challenges and identifying future possibilities for integration. The review focuses on two key areas:

1. First, it explores ML algorithms and techniques used in conjunction with CFD simulations in UGI research, a relatively underexplored area. While some studies have acknowledged the high performance of CFD-ML integration [47,48], there is still a need for a more in-depth understanding of how ML technology can be applied specifically to UGI research. This includes a thorough review of the different ML algorithms employed, their effectiveness in supporting CFD simulations, and their potential for enhancing predictions related to urban heat and air quality.
2. Second, the study highlights the challenges limiting the broader adoption of CFD-ML integration and offers recommendations for future research. By addressing these obstacles, the review aims to promote the wider use of these technologies in urban environmental planning and analysis.

The review is structured as follows: Section 2 outlines the methodology, including literature search and information extraction. Section 3 explores the mechanisms through which vegetation and urban water bodies influence urban environments, and evaluates CFD modeling methods and available tools. Section 4 focuses on the integration of CFD and ML in UGI research, covering topics such as algorithm classification, integration methods, data requirements, and validation approaches. Section 5 examines specific case studies that apply integrated CFD-ML methods in UGI research, providing a summary of the methodologies and application areas. Section 6 offers a detailed analysis of key findings from various perspectives, interpreting results, identifying limitations, uncovering potential causal relationships, and exploring the broader implications for the field. Finally, Section 7 summarizes conclusions, emphasizing theoretical and practical significance, and proposes future directions for integrating ML and CFD in UGI research.

2. Method

This paper conducts a systematic review by defining the research questions, objectives, and incorporating relevant literature sources. The key research questions addressed in this study are: (1) *What are the simulation strategies for CFD in UGI research?* (2) *How does ML enhance CFD simulations in urban and UGI design, and what are the key characteristics of ML?* (3) *What are the available integration strategies between ML and CFD?* (4) *How can the integration of ML and CFD advance studies on heat mitigation and air quality improvement in UGI?*

The study assesses the impact of UGI on urban thermal environments and air quality, focusing on advanced techniques, particularly the integration of ML and CFD. A systematic literature review was conducted using the PRISMA method [50] to capture current developments, methodologies, and research gaps in CFD-ML integration for UGI studies. The review process is divided into three stages: planning, conducting, and reporting [45].

During the planning stage, study objectives, relevant keywords, and clear inclusion/exclusion criteria were established to ensure that the review would offer valuable insights into the opportunities and challenges of applying CFD and ML in the context of UGI. Comprehensive searches were conducted in databases such as Web of Science, Google Scholar, Wiley Online Library, and Scopus, including books, journal articles, and conference papers, using keywords detailed in Fig. 3. Keywords were applied in two distinct search strategies: the first excluded ML terms to focus solely on CFD applications for urban heat mitigation and air quality; the second replaced UGI-specific keywords to broadly explore CFD and ML integration in urban contexts. This dual approach was designed to enrich the understanding and application of CFD-ML integration in UGI design. The search queries were conducted

on March 1, 2025. After review, 146 studies were selected, supplemented by 31 manually added studies, totaling 177 sources.

With the rapid growth in theoretical and applied research, it has become increasingly challenging to conduct comprehensive surveys of early-stage studies, systematically organise extensive amounts of information, and effectively propose future research directions. To address this challenge, this study provides a contemporary integrated methodological perspective grounded in keyword distribution analysis and technical categorisation aligned with the ML era.

As shown in Fig. 4, this review utilises the software tool VOSviewer to perform a visual analysis of keywords extracted from the literature, aiming to clarify and evaluate research patterns within the relevant

field. Each node (circle) represents a keyword or term within the dataset in the figure. The size of each node corresponds to the frequency of that keyword's occurrence in the literature, with larger nodes indicating higher frequencies. The thickness of the lines connecting nodes represents the strength of association between keywords, determined by how frequently two keywords co-occur within the same publication; thicker lines indicate stronger co-occurrence relationships.

Within the analysed dataset, the most frequently appearing keywords are "construction and building technology" and "environment science & ecology," followed by "microclimate," "pollutant dispersion," and "vegetation." Among methodological keywords, "CFD" and "Machine Learning" are the most frequently used terms. The software tool "ENVI-

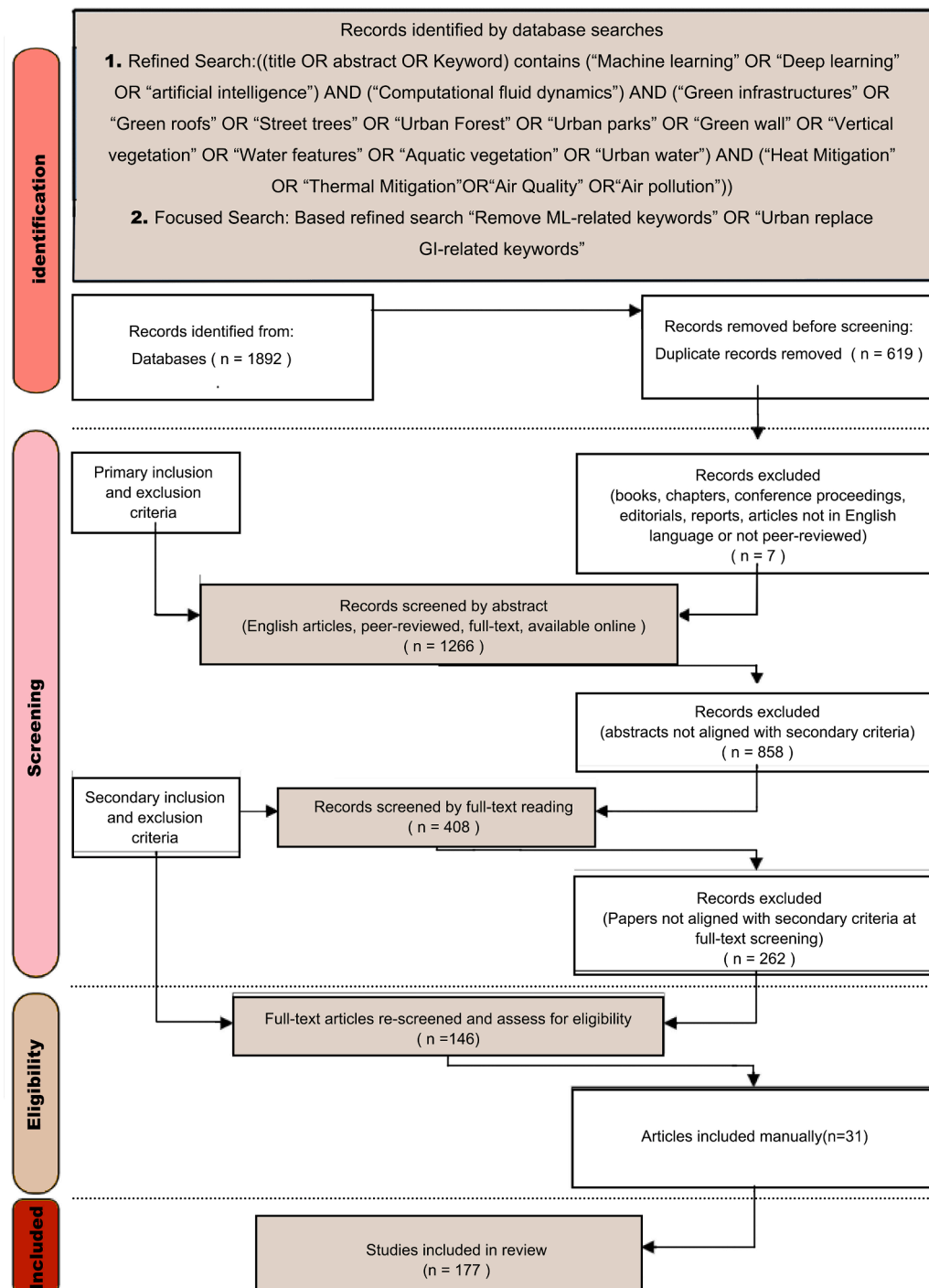


Fig. 3. Steps of the PRISMA method.



Based on these keywords cloud map the review is structured into four thematic categories, as follows:

- Thermal environment (thermal comfort, urban heat island, overheating)
- Air pollution (air quality, particle size, PM_{2.5})

- CFD numerical simulation (controlled study, accuracy, large-eddy simulation, RANS, turbulence, time step)
- Purpose of simulation (optimization, mitigation, urban green design, cooling effects, path design)
- CFD software (ENVI-met, OpenFOAM)

- UGI types (green roofs, trees, forestry, water, green walls)

- ML algorithms (Random forest, artificial neural network, genetic programming, deep learning, Fourier neural operator)
- ML tools and integration technology (PyTorch, dimensionality reduction, super-resolution, ML-guided adaptive meshing)
- Function (prediction, high-performance computing, computational efficiency, cost reduction, automation)

This section provides an overview of CFD simulations of UGI (Section 3.1), the cooling mechanisms of UGI (Section 3.2), the impact of UGI on air quality (Section 3.3), and the geometric parameters of vegetation and water bodies within UGI (Section 3.4). Additionally, Section 3.5 presents a comprehensive review of available CFD tools, with a particular focus

on microclimate simulation studies influenced by UGI.

3.1. CFD workflow and simulation techniques in UGI research

When integrating CFD and ML for UGI research, understanding the CFD workflow is crucial for structuring fluid dynamics solutions and identifying ML's role in optimizing processes, reducing costs, improving accuracy, and streamlining tasks like mesh generation and turbulence modeling. This ensures ML results remain reliable and consistent with physical principles.

CFD is highly effective for simulating microscale physical processes in urban structures by solving governing equations for mass, momentum, and thermal energy conservation. It incorporates turbulence models, chemical species transport equations, and can handle single-phase or multiphase flow phenomena, making it a powerful tool for urban airflow, pollutant dispersion, and thermal comfort studies [44]. CFD can resolve turbulence using models such as Reynolds-Averaged Navier-Stokes (RANS) and Large Eddy Simulation (LES), typically achieving meter-scale spatial resolution in urban canopy simulations [51]. This enables a detailed analysis of microscale atmospheric processes within cities. However, CFD alone does not fully capture meso-scale atmospheric dynamics and is often coupled with larger-scale atmospheric models for improved accuracy. It provides a structured method for addressing urban fluid dynamics challenges.

Fig. 5 illustrates the CFD workflow, outlining critical steps in UGI research. The first step involves vegetation modeling, which can be performed either by explicitly resolving plant geometry or by employing parameterized (“porous”) models that introduce source terms into governing equations. Common approaches include explicit geometric models, simplified bulk geometry, and porous media (volume-averaged) models. Explicit geometric models entail detailed vegetation representations (e.g., resolving individual leaves or realistic canopy structures) and cover fully detailed three-dimensional plant models [44]. Although these explicit models accurately capture microscale flow phenomena near vegetation, they substantially increase mesh complexity and computational cost, typically limiting their use to small-scale analyses.

Simplified bulk geometry models approximate vegetation canopies with basic geometric volumes (e.g., ellipsoids [34,52], cylinders [53], hexahedrons [54], and cones [55]), capturing wind blockage and shading effects without requiring extensive internal mesh refinement. However, these models do not fully resolve internal porosity, potentially leading to overestimated aerodynamic resistance without proper calibration [56]. Porous media (volume-averaged) models, commonly used in CFD vegetation modeling, represent vegetation as porous regions by including momentum and turbulence source terms in standard fluid equations, thereby achieving a balance between realism and computational efficiency [43,57]. These effects are typically implemented through leaf area density (LAD) profiles and drag coefficients, producing distributed momentum sinks and additional turbulence dissipation [44]. Pollutant removal through vegetation can also be modeled as volumetric sinks in species transport equations (with deposition rates proportional to LAD), and evaporative cooling effects are introduced as volumetric cooling source terms [45,47].

Next, the computational domain is defined according to research objectives. Simulations may use tools like ENVI-met to customize domain sizes based on urban layouts or replicate wind tunnel experiments for controlled validation. Domain sizes generally follow established best practices [58,59]. Typically, side boundaries are usually treated as symmetry planes, while the top is often treated as a slip/symmetry plane. Guidelines suggest setting distances from the model to side symmetry walls at 5H (where H is the model height), and from the model to the top, inlet, and outlet boundaries at 5H, 5H, and 15H, respectively, ensuring accurate flow dynamics representation without boundary interference. Boundary conditions are then established. In CFD simulations of urban environments [60], wind profiles [61] are typically applied at inlet boundaries, defining parameters like velocity, turbulent kinetic energy, and ϵ . In some cases, inlet wind profiles are derived from wind tunnel experiments [21] or field test measurements. Beyond wind profiles, other crucial boundary conditions must be set. Wall boundary conditions are applied to building surfaces, affecting airflow and turbulence.

Accurate mesh generation is critical in CFD simulations involving

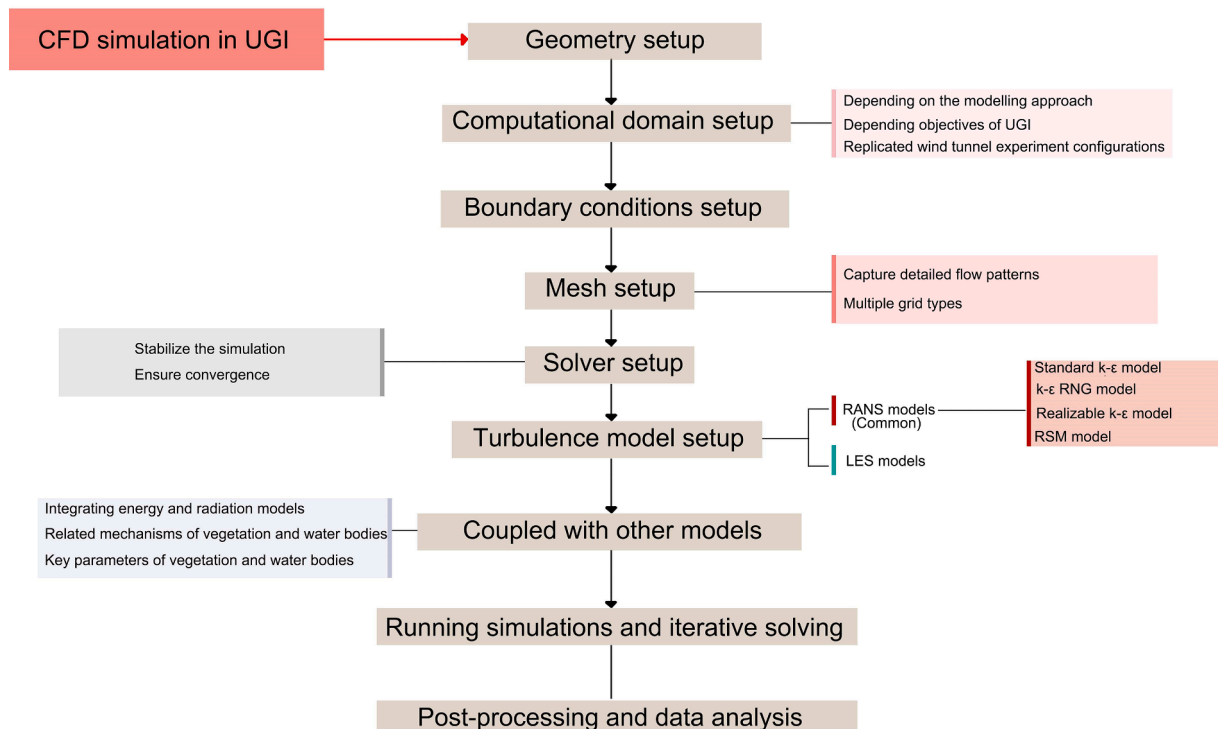


Fig. 5. CFD modeling workflow for UGI.

complex urban geometries and vegetation structures, as it subdivides the computational domain into finite volumes for numerical solution of fluid dynamics equations, ensuring simulation accuracy. The finite volume method (FVM) is widely employed to discretize governing equations, accurately ensuring conservation of mass, momentum, and energy [62]. Structured hexahedral, tetrahedral, and polyhedral grids are common mesh types, with hybrid meshes often adopted in complex urban environments to optimize resolution and computational efficiency [63]. Typically, structured hexahedral or prismatic grids are applied near walls and ground surfaces, while unstructured tetrahedral/polyhedral grids fill regions characterized by complex building shapes and vegetation forms, with finer mesh resolution used selectively in regions of interest [17,22,58]. Employing porous media models simplifies meshing by eliminating the need to explicitly resolve individual leaves or branches, significantly reducing computational complexity. In contrast, explicitly representing vegetation as detailed solid objects (e.g., small cylinders or leaf surfaces) greatly complicates mesh generation and substantially increases cell counts. Finally, to ensure the reliability of CFD results, mesh sensitivity analyses, often utilizing the grid convergence index (GCI) [52], verify that mesh resolution does not significantly affect the simulation outcomes.

Choosing an appropriate turbulence model is vital for accurately predicting turbulent flows, heat transfer, and pollutant dispersion. RANS and LES models are commonly employed in UGI simulations. The RANS model, especially the k - ϵ scheme, is widely used in tools like ANSYS Fluent, OpenFOAM, and ENVI-met [64]. It is particularly effective for simulating the effects of vegetation, water bodies, and solar radiation due to its computational efficiency under typical conditions. RANS models are well-suited for handling many complex flow situations and turbulence [65,66]. By averaging velocity fields over time, RANS simplifies turbulent flows, reducing computational costs while maintaining reasonable accuracy for practical applications. The standard k - ϵ model, known for its simplicity and efficiency, is ideal for routine applications like preliminary airflow assessments around basic vegetation layouts or water bodies [67]. For more complex and anisotropic flows, enhanced models such as the k - ϵ RNG [68], Realizable k - ϵ [69], and Reynolds Stress Model [70] provide greater accuracy, better capturing complex turbulence structures typical in urban green infrastructure, especially when precision is essential.

In contrast, LES achieves higher precision by resolving large turbulent structures while modeling smaller scales through subgrid-scale models, offering more accurate predictions of turbulent flows and pollutant dispersion compared to RANS [71]. The method is computationally demanding because it needs very fine spatial grids and small time-steps to capture those eddies; generating realistic, time-dependent turbulent inflow profiles and other boundary conditions adds further overhead [43]. Additionally, LES requires advanced subgrid-scale models and carefully specified time-dependent inlet and boundary conditions, increasing simulation complexity [72].

3.2. Cooling mechanisms of UGI and their role in heat mitigation

Building on the CFD modeling discussion in Section 3.1, understanding the specific impact mechanisms of UGI is essential for accurately predicting its role in urban heat mitigation. Vegetation cools urban environments through various physical processes, including radiation, shading, convection, and transpiration, driven by vegetation's physical properties and its interactions with environmental factors. As outlined in Table 1, the five primary types of UGI exhibit distinct thermal mechanisms, with certain processes dominant depending on UGI type and characteristics.

The primary contributors to ground cooling are URP, ST, and UW. URP stand out due to their significant cooling potential. Their diverse vegetation species and large area coverage allow URPs to utilize nearly all common cooling mechanisms [73]. These include the shading effect of trees, which reduces direct solar heat absorption, and transpiration,

which enhances latent heat release from plants [69]. These processes collectively lower local air temperatures, making URPs a key component of urban cooling strategies. Additionally, the cool air produced within the park is dispersed to surrounding areas through diffusion, advection, and convection, further amplifying its cooling effect [74]. ST provides targeted shading and transpiration cooling for streets and sidewalks, as well as altering airflow to enhance local cooling [75] although dense crowns can also impede ventilation under some street-canyon geometries.

For UW, the key cooling mechanisms include evaporation, thermal capacity, and airflow modification. The high thermal capacity of water, combined with its ability to cool surrounding areas through evaporation and heat exchange [63], makes urban water bodies a crucial factor in regulating the urban microclimate. This includes common types such as rivers, lakes, and artificial waterways.

For rooftops and building facades, GW offers a vertical cooling solution. These structures incorporate vertical plant growth, supported by frames, growing mediums, and irrigation systems, to reduce the temperature of walls and their surrounding air. The combined effects of shading, transpiration, and evaporation in green walls are particularly effective in cooling the surfaces they cover, helping mitigate urban heat island effects by lowering surface and ambient temperatures [76,77]. Similarly, GR provides cooling potential through transpiration, evaporation, and shading. By including multiple layers such as a support structure, substrate, and plant canopy, green roofs help reduce building surface temperatures, thereby decreasing indoor cooling demands. The interaction between these thermal processes leads to enhanced heat mitigation in urban environments.

3.3. Mechanisms of UGI for air quality improvement

UGI simultaneously contributes to urban cooling and air-quality improvement through tightly coupled thermodynamic and chemical processes. Recent empirical evidence at multiple spatial scales quantitatively demonstrates this synergy. At the local scale, field measurements from 108 plant communities in Beijing revealed that a moderate decrease of approximately 0.8 °C (-2.7 %) in canopy air temperature corresponded to an average 10.6 % increase in PM_{10} retention efficiency that is, the vegetation physically trapped airborne PM_{10} particles on leaf surfaces; notably, 90.9% of these communities exhibited simultaneous cooling and particulate removal benefits [85]. At the regional scale, long-term monitoring at Belsk, Poland (1995–2023) indicated that average daytime ozone concentrations (67.9–86.8 $\mu\text{g}/\text{m}^3$) increased with rising temperatures (15.9–20.3 °C), with an estimated slope of $1.56 \pm 0.87 \mu\text{g}/\text{m}^3/\text{°C}$, confirming a positive ozone-temperature relationship [86]. Beyond temperature-driven effects, evapotranspiration significantly influences $PM_{2.5}$ dry deposition. An i-Tree Eco assessment for Shanghai identified a positive correlation ($r = 0.52$, $p < 0.01$) between cumulative evapotranspiration and $PM_{2.5}$ removal, with maximum deposition ($\approx 0.61 \text{ g m}^{-2}$) occurring at moderate evapotranspiration levels (38–42 mm per 8 days). Deposition efficiency declined when evapotranspiration exceeded 50 mm, likely due to partial stomatal closure limiting particle capture [87]. Collectively, these multi-scale findings provide robust evidence that heat mitigation and air-quality improvement by UGI are fundamentally interdependent, measurable, and synergistically linked, underscoring the importance of integrative strategies in urban environmental management.

Besides influencing air quality through thermal pathways, UGI directly improves air quality via mechanisms such as pollutant deposition, absorption, resuspension, and wind flow modification (Table 2). However, pollutant removal efficiencies differ significantly by pollutant type and UGI form.

For particulate matter (e.g., PM_{10} and $PM_{2.5}$), URP and ST primarily facilitate the deposition and inertial impaction of coarse particles through their canopy structures, while fine particle removal relies on diffusion enhanced by leaf micro-roughness [91,96]. GW efficiently

Table 1
The mechanisms of UGI in heat mitigation.

UGI	Heat mitigation mechanisms							Refs.
	Radiation	Shading	Transpiration	Evaporation	Wind flow modification	Thermal capacity	Key features	
URP	☑	☑	☑	☑	☑	☑	Extensive green and blue areas form park cool islands, dispersing cool air through diffusion, advection, and convection.	[69,73,74]
ST	☑	☑	☑		☑		Reduces surface temperatures of streets and sidewalks, but has no significant cooling effect on rooftops or upper levels.	[66,75,78]
GW	☑		☑	☑	☑	☑	Vertical plant structures cool walls and surrounding air.	[76,79–81]
GR	☑		☑	☑	☑	☑	Reduces building surface temperature through layered support and thermal interactions.	[77,82]
UW	☑			☑	☑	☑	High thermal capacity and evaporation effect from water bodies (rivers, lakes, artificial waterways).	[63,83,84]

Abbreviations: urban forests and parks (URP), street trees (ST), green walls (GW), green roofs (GR), urban water bodies (UW).

filter coarse particles due to vertical leaf surfaces; vegetation with higher surface roughness achieves greater deposition velocities [99]. For GR, particle removal is limited by their elevated positioning [101]. UW effectively remove coarse PM through wet deposition, though fine particle removal via weak Brownian diffusion is limited but enhanced under onshore breeze conditions inducing downdrafts over water surfaces [84].

For gaseous pollutants such as VOCs, dry deposition is a key removal mechanism, particularly under high background concentrations, and is enhanced by increasing leaf area density and favorable exposure to wind flow. By contrast, nitrogen oxides (NO_x) exhibit much lower dry-deposition velocities, so atmospheric dispersion remains the primary control on their near-surface concentrations [89]. GW exhibits limited stomatal uptake due to frequent water stress and shorter leaf lifespan [100]. GR can absorb gaseous pollutants, but their removal efficiency is typically lower than at tree-canopy height due to dilution at rooftop level [102]. Urban water bodies dissolve soluble gases, but efficiency depends on surface turbulence intensity [49].

Regarding wind flow modification, tree morphology in URP and ST promotes pollutant dispersion through diffusion, advection, and convection, thus improving air quality in adjacent urban areas [90,97]. However, densely planted trees may form recirculation zones, trapping pollutants near the surface. GW and GR exert limited aerodynamic influence on urban ventilation due to their low roughness. Notably, UW induce local airflow driven by thermal contrasts due to their high heat capacity, thereby diluting pollutant plumes in nearby areas [71].

3.4. UGI parameters in CFD simulation

Following the detailed discussion of UGI's impact on heat mitigation and air quality improvement, accurately modeling these effects in CFD simulations requires careful consideration of key parameters, crucial for generating reliable training data for ML algorithms and analyzing parametric effects. The effectiveness of UGI depends inherently on its

specific characteristics, which must be accurately reflected in simulations. Essential parameters include vegetation type, structure, location, and unique vegetation features influencing heat mitigation and pollutant dispersion. Precisely incorporating these parameters ensures more reliable predictions in CFD and ML models, enhancing the accuracy of urban environmental assessments.

As outlined in Table 3, these parameters serve as critical inputs for simulating the interactions between UGI and urban microclimates. For instance, Li et al. [103] analyzed tree geometry to construct vegetation models in two steps: first, defining basic morphology, such as tree height, maximum LAD, and canopy height; then specifying detailed leaf and root characteristics based on LAD and leaf area index (LAI). LAD refers to the total one-sided leaf area per unit of canopy volume [100], and LAI represents the ratio of leaf area to ground area [104]. While LAI provides a macro-level understanding of vegetation coverage, LAD offers deeper insights into vegetation structure; both are crucial for accurately modeling airflow resistance and UGI environmental impact [52,73].

Tree age also significantly impacts CFD modeling, especially for URP and ST. As trees mature, LAD and LAI change, influencing heat dissipation and air circulation. Li et al. [105] found that trees aged 30–60 years offer optimal heat dissipation, while those older than 60 may obstruct airflow, reducing pollutant diffusion and limiting heat mitigation.

For URP, large coverage areas offer a distinctive cooling strategy, enabling cool air diffusion across considerable distances. A simulation of Singapore showed cooling intensity decreasing by 0.6 °C at 117 m from the park [106]. Similarly, in GR scenarios, building height significantly affects heat dissipation. Jia and Wang [64] suggest that green roofs at lower heights have a more pronounced cooling effect at the pedestrian level.

Regarding UW, its configuration and positioning significantly influence heat mitigation and air quality. Larger water surfaces enhance cooling effects under prevailing winds, while static or dynamic flow

Table 2
Air quality impact mechanisms of UGI types.

UGI	Air quality impact mechanisms				Key features	Refs.
	Deposition	Absorption	Resuspension	Wind flow modification		
URP	☑	☑		☑	Offer diverse vegetation promoting pollutant deposition and absorption, modifying wind flow to enhance air quality.	[88–91]
ST	☑	☑	☑	☑	Capture dust and gases from traffic but may have mixed effects due to altered wind flow.	[92–98]
GW	☑	☑		☑	Support pollutant deposition, absorption, and reduce wind speeds.	[99,100]
GR	☑	☑		☑	Aid particulate deposition, modify wind patterns, potentially improving air quality depending on their height and arrangement.	[101,102]
UW	☑	☑		☑	Promote pollutant deposition on surfaces and affect wind flow, facilitating pollutant dispersion.	[49,71,84]

Abbreviations: urban forests and parks (URP), street trees (ST), green walls (GW), green roofs (GR), urban water bodies (UW).

Table 3
Different UGI parameters in selected CFD simulation studies.

Refs.	UGI	Parameters	Key findings
[108]	URP	The arrangement of vegetation, vegetation type, tree age, height of tree, crown diameter, LAD, LAI.	Larger canopies and leaf areas provide greater wind speed reduction.
[109]	URP	Surface roughness of vegetation, the areas covered by grass, LAD.	Tree species with high LAD and strong deposition characteristics enhance deposition and aerodynamic dispersion.
[105]	ST	Tree size, height of tree, crown diameter, LAD, LAI, the leaf energy balance, tree age, the minimal stomata resistance of the leaves, the albedo of the leaves.	At pedestrian level, trees younger than 30 years have limited cooling capacity; older trees significantly contribute through shading and transpiration cooling.
[57]	GW	Morphological plant, leaf shape, leaf size, plant area density, porosity, leaf ratio, specific leaf area, leaf dissection index, functional leaf size..	A Darcy-Forchheimer model characterizes airflow through vegetation reliably at low pressure drops. Vegetation permeability, Forchheimer-drag, and porosity serve as CFD modeling inputs.
[110]	GW	Vegetation type, LAD, LAI, air gap between substrate and wall, the overall leaf area, spacing, height, and width of the vertical plants, substrate for plant, substrate and wall.	Keeping total leaf area constant, increasing pedestrian-level GW coverage may improve thermal comfort in urban street canyons.
[82]	GR	LAI, height of plants, leaf reflectivity, minimum stomatal resistance, the substrate layer components.	Heat mitigation of green roofs depends on climate conditions, orientation, slope, LAI, substrate depth, moisture content, and plant species.
[83]	UW	The water surface area size, depth, the water body's type, water body's temperature	Ventilation performance of waterways and roads depends on their orientation and width; open space ventilation is influenced by location; vegetation area ventilation is determined by porosity.
[49]	UW	The flow characteristics of water bodies, water body shape.	In areas with steady wind direction, elongated waterbodies aligned with prevailing winds enhance cooling intensity.

Abbreviations: urban forests and parks (URP), street trees (ST), green walls (GW), green roofs (GR), urban water bodies (UW), leaf area density (LAD), leaf area index (LAI).

characteristics impact heat exchange and pollutant deposition, as seen between ponds and rivers [49]. The shape of water bodies also matters, with more complex profiles creating greater contact with the surrounding environment, enhancing cooling [65]. Moreover, turbulence intensity above water affects pollutant dispersion/dilution, highlighting water bodies' role in air quality regulation by promoting vertical and horizontal pollutant mixing [107].

3.5. Available CFD tools and potential integration with ML

Effective integration of CFD models relies on advanced CFD tools. Table 4 summarizes several frequently used CFD tools for simulating UGI's impact on heat mitigation and air quality. Among these, ENVI-met and OpenFOAM are most commonly cited for their effectiveness in modeling vegetation and urban environments. Specifically, ENVI-met is preferred due to its extensive vegetation library detailing plant density and structure, enabling precise simulation of vegetation-microclimate interactions and accurate representation of key UGI parameters such as LAD and LAI discussed in Section 3.4.

In a study by Liu et al. [112], ENVI-met's vegetation model database streamlined simulations by enabling rapid input of vegetation

characteristics, enhancing its application in UGI research. Another widely used tool, OpenFOAM, integrates flexibly with Rhino and Grasshopper plugins for advanced simulations. For example, OpenFOAM combined with point cloud data effectively analyzes vegetation's impact on urban microclimates [106]. Moreover, the Grasshopper plugin supports various programming languages and allows genetic algorithms to be linked using Python, providing an innovative workflow to optimize UGI design and urban cooling strategies [125].

OpenFOAM and Rhinoceros offer high integration potential due to their open-source nature and flexibility, supporting ML-driven optimizations in mesh generation and flow prediction [115]. Tools such as ENVI-met and ANSYS Fluent have moderate integration potential, usually requiring external customization for ML tasks like parameter tuning and surrogate modeling [36,119]. In contrast, COMSOL and PHOENICS show limited integration possibilities, restricting ML applications mainly to external data processing. Therefore, these differences highlight the importance of selecting tools that support ML integration. In the following section, we will explore how coupling CFD with ML-based solutions can help optimize UGI simulations.

4. Integration of CFD and ML technology

4.1. Machine learning algorithms overview

ML focuses on developing algorithms that can learn from data, emulating human learning processes to enhance accuracy and efficiency [127]. In the context of CFD, ML offers significant potential for improving simulation speed and quality. As detailed in Section 3.1, high-fidelity urban-scale CFD requires carefully designed hybrid meshes and, where appropriate, adaptive refinement; these fine grids incur large computational costs, especially when advanced turbulence models are used. ML-based algorithms help alleviate these computational demands by replacing or supplementing traditional numerical methods with empirical, data-driven approaches [128].

Transitioning from purely physics-based CFD to integrating ML represents a complementary approach, allowing ML to efficiently handle large datasets and identify patterns. In UGI research, ML is particularly useful for analyzing extensive environmental data, predicting outcomes, and uncovering subtle trends. This section reviews ML algorithms enhancing UGI understanding, design, and management, providing a framework for future CFD integration to achieve efficient, comprehensive solutions.

Fig. 6 presents ML algorithms for evaluating urban thermal environments and air quality, categorized into tree-based, kernel-based, neural networks, regression, and others. Neural network-based approaches are the most commonly used due to their flexibility and powerful ability to model complex patterns. Among these, the multi-layer perceptron (MLP), one of the simplest forms of neural networks, remains a core tool in machine learning for its ease of use and effectiveness across a wide range of classification tasks. The adaptability of MLPs to various types of data and problems highlights the versatility of neural networks in addressing diverse challenges in environmental science. Tree-based methods are the second most popular, known for their performance in both prediction and classification tasks. This group of algorithms excels at handling high-dimensional data and capturing nonlinear relationships, often without the need for extensive data pre-processing [36,127]. Overall, ML applications have been used slightly more for air quality research over heat mitigation.

The classification of these algorithms is largely based on their grouping in existing literature, with tree-based, kernel-based, regression, neural networks, and others categorized as supervised learning, relying on labeled data for training. Supervised Learning is especially effective in urban research because it can analyze specific inputs and predict outcomes with high accuracy. Its ability to incorporate diverse metrology parameters allows for the prediction of variables like city air quality, particularly when large labeled datasets are available [129].

Table 4
Overview of CFD tools and typical applications.

CFD tools	Description	UGI types	Key functions	Limitations	Refs.
ENVI-met	A 3D urban microscale modeling software with various plant modules.	Street trees, green roof, green wall, parks, urban water	Simulates outdoor thermal comfort, air pollution analysis, and vegetation characteristics.	Limited for large-scale or long-term studies; Low interoperability with other tools.	[36,49, 111–113]
OpenFOAM	An open source programmable (C++) multi-physics CFD toolbox	Street trees, parks, Urban water	Provides multiple CFD model options with a wide range of numerical solvers and high programmability.	Complex mesh generation; High learning curve; No pre-built UGI models.	[104,106, 109,114, 115]
ANSYS Fluent	A comprehensive CFD software designed for high-fidelity fluid dynamics simulations.	Street trees, green roof, green wall, parks, urban water	High accuracy with the finite volume method; supports large-scale, high-precision simulations.	Requires advanced expertise; Long computation times for large-scale simulations.	[57, 116–119]
STAR-CCM+	A multiphysics simulation software	Street trees, green roof, green wall	Calculates air pollution concentrations and aerodynamic effects of porous media.	High computational costs for detailed simulations; Limited UGI-specific libraries.	[60,120]
CityFFD	A new CFD model for fast and accurate simulation		Uses Large Eddy Simulation (LES) method; suitable for rapid simulations of urban environments.	Lacks vegetation and pollutant models; Simplified fluid dynamics; Limited environmental physics.	[121,122]
COMSOL	A CFD software for coupling multiphysics and single-physics modeling.	Green wall	Models vegetation porosity and other environmental interactions.	Limited for complex urban flows; k-ε model may underestimate recirculation.	[57]
PHOENICS	A cost-effective CFD program.	Green roof, green wall, street trees	Models heat and mass transfer, fluid flow, and the effects of vegetation.	Low automation in mesh generation; Lacks advanced visualization options.	[52,101, 123]
Rhinoceros	A 3D building modeling software includes CFD plugins and Python scripts.	Street trees, parks, Urban water (OpenFOAM)	Allows multi-objective and coupled simulations; often used with OpenFOAM for urban modeling.	Basic fluid modeling; Limited environmental modeling compared to specialized CFD tools.	[124,125]
uDALES	A large-eddy simulation software for urban flow, dispersion and microclimate.	Street trees	Models complex urban morphologies and non-neutral boundary layers with a focus on LES simulations.	High computational resources needed; Complex setup and validation; Limited built-in vegetation models.	[126]

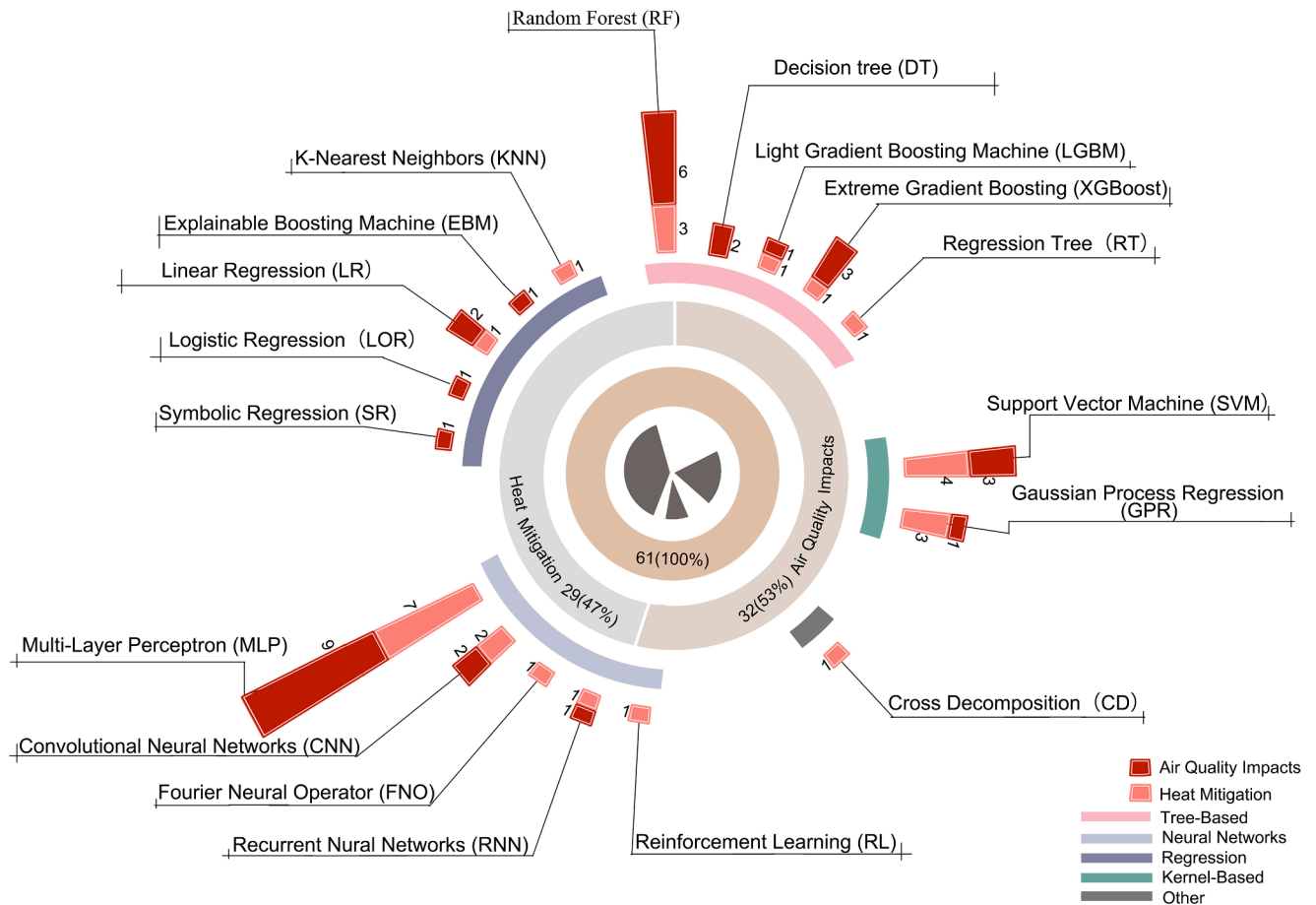


Fig. 6. A circular bar chart of algorithm applications.

Conversely, unsupervised learning utilizes unlabeled data, avoiding costly labeling processes associated with supervised methods [130], which is a significant challenge in Supervised Learning [128]. The main

objective of unsupervised Learning is to identify patterns and simplify data, allowing for efficient problem analysis and trend detection in urban environments [131].

Most general-purpose algorithms are neural networks that consist of multiple layers of neurons. They can adapt to both supervised and unsupervised learning by adjusting the number of layers and neurons in each layer [128]. This flexibility makes neural networks a popular choice in applications for both urban heat mitigation and air quality improvement. Neural networks' ability to handle large, complex datasets and identify subtle patterns makes them effective across diverse challenges in urban research.

As machine learning continues to gain traction in urban research, the increased availability of open-source datasets and pre-trained models will likely drive further adoption of supervised learning. It is predominantly used for classification and regression tasks, both of which align with the goals of research in urban heat mitigation and air quality improvement. Classification models are particularly valuable for flow-regime categorisation or design-option screening [132], while regression models offer predictive insights into pollutant concentrations across different urban areas [129].

4.2. Approaches for integrating ML and CFD

This section focuses on the integration of ML and CFD in urban studies, particularly for analyzing the role UGI in heat mitigation and air quality improvement. The combination of ML and CFD accelerates the modeling process and enhances predictive capabilities, enabling more dynamic and responsive urban design strategies for UGI. For further details on mesh generation and solver set-up in the underlying CFD workflow, readers are referred to Section 3.1. The remainder of this section therefore focuses exclusively on coupling strategies and ML integration methods.

Traditionally, the integration of ML and CFD can be broadly categorized into two approaches: Coupled CFD-ML and Surrogate CFD-ML (Table 5). In Coupled CFD-ML, ML models function as auxiliary tools for conventional CFD frameworks by incorporating ML into the simulation process, most notably for tasks such as turbulence modeling and flow control (Fig. 7a). For example, deep reinforcement learning (DRL) has been applied to solve complex flow control and optimization problems in fluid dynamics, and has also been used to predict microclimates and building energy consumption when combined with CFD [133]. Tools such as OpenFOAM and ANSYS Fluent, combined with various ML models, have demonstrated enhanced capabilities in optimizing urban environments and improving pedestrian comfort or pollutant concentration predictions. For instance, deep learning models combined with ENVI-met have been used to assess both indoor and outdoor thermal performance, while the backpropagation neural network (BPNN) has shown high accuracy in traffic-related pollution predictions.

On the other hand, Surrogate CFD-ML involves employing ML models as a complete replacement for traditional CFD simulations, where ML is used to construct surrogate models or response surfaces to approximate the behavior of the original CFD solution (Fig. 7b). For example, surrogate models like the BPNN or random forest (RF) have increased the speed of wind speed and pollutant concentration predictions, with performance improvements of up to 800 times faster compared to traditional CFD approaches. Moreover, neural networks, SVM, and regression models have been used in surrogate CFD-ML to estimate air quality and thermal comfort in large-scale urban environments.

As emphasised in Section 3.1, once a mesh independent CFD baseline has been established, the most crucial stage in both methods is the calculation of environmental indicators produced by CFD simulations. In surrogate CFD-ML, the accuracy of the CFD data used for training directly impacts the performance of the ML model. In contrast, coupled CFD-ML involves running CFD and ML models in parallel, with the data from CFD simulations combined with ML data for enhanced analysis. The amount of training data is a key factor in ML performance. As shown in Table 5, some studies required as many as 1000 training cases. In general, the larger the dataset, the better the prediction accuracy of ML

models [37]. However, this does not imply that more data always leads to better results. For example, one case study predicting urban wind speed found that accuracy plateaued once the dataset reached 80,000 samples [115]. This demonstrates that the optimal dataset size depends on the specifics of each research project.

To ensure the reliability of ML and CFD integration, validation is typically carried out using reliable datasets. In both Coupled and Surrogate CFD-ML, results are often compared with data from wind tunnel experiments or CFD simulations. For Surrogate CFD-ML, validation can be more straightforward since only a small portion of the CFD data needs to be set aside as test data. For instance, in a pedestrian wind study, 150 CFD simulations were used as training data, while the remaining CFD simulations served as validation data [134]. However, wind tunnel testing remains one of the most reliable methods, as it replicates wind conditions around buildings, making it a valuable benchmark for validation [20]. In the next sections, we explore how dimensionality reduction methods accelerate CFD computations, how deep reinforcement learning optimizes CFD simulation control, how machine learning can enhance adaptive mesh refinement, and other emerging CFD-ML technologies, all with significant potential to advance UGI design.

4.3. Dimensionality reduction techniques in CFD-ML

High-fidelity CFD simulations are computationally intensive, especially when numerous parametric iterations, e.g., in UGI design optimization, are required. This has motivated researchers to adopt dimensionality reduction techniques (Fig. 8) to construct Reduced-Order Models (ROMs), thus reducing computational costs [150]. By extracting the dominant modes of a flow field, these methods map the original high-dimensional problem onto a low-dimensional subspace, significantly lowering the number of degrees of freedom. For example, proper orthogonal decomposition (POD), often regarded as principal component analysis applied to flow fields, is commonly employed to identify the most significant modes and build approximate models based on them [151]. Compared with direct CFD computations, such ROMs can greatly accelerate design iterations for UGI components, such as optimal tree placement or water-body configuration, while retaining essential flow characteristics [152].

A typical example involves combining POD with interpolation or machine learning to predict the flow-field coefficients, enabling the rapid estimation of aerodynamic forces and flow behavior [153]. The key advantage of linear decomposition methods like POD is their interpretability, as modes correspond to coherent physical structures (e.g., primary vortices, thermal plumes) [154], greatly facilitating understanding of how specific UGI elements, such as tree canopies or green walls, influence airflow. However, linear methods such as POD may inadequately capture complex, nonlinear interactions prevalent in UGI scenarios involving heterogeneous vegetation-induced turbulence. To address this, nonlinear dimensionality reduction methods have been introduced.

Kernel PCA, extending PCA by mapping data into a high-dimensional feature space via kernel functions before performing PCA, effectively captures nonlinear, curved manifolds otherwise missed by linear methods [155]. Despite its theoretical superiority in representing nonlinear complexities, kernel PCA faces significant scalability challenges, requiring computation and storage of an $n \times n$ Gram matrix for n CFD snapshots, consuming $O(n^2)$ memory and $O(n^3)$ computational time [156]. Additionally, its sensitivity to kernel choice and hyperparameter tuning significantly impacts performance and practical applicability [157].

Autoencoders, neural networks trained for nonlinear data compression and reconstruction, effectively encode complex CFD fields into a compact latent space, enabling efficient reconstruction via convolutional architectures. This approach, classified as Non-Intrusive Reduced-Order Modeling (NIROM), bypasses the integration with CFD solvers, learning directly from data [158]. Studies on urban pollutant dispersion

Table 5

Examples of tools and algorithms employed in ML and CFD integration for heat mitigation and air quality enhancement.

Coupled CFD-ML						
ML algorithm	CFD model	CFD tools	Training and test data	Summary/ Key findings	Refs.	
MLP: DRL	RANS	OpenFOAM	CFD and ML are parallel	Open-source platform DRLinFluids significantly reduces the workload of DRL in CFD.	[134]	
MLP: Artificial Neural Network (ANN)	Realizable k- ϵ turbulence model	ANSYS Fluent	150 Architectural design variables	The framework optimizes pedestrian comfort in heat and cold conditions.	[135]	
MLP		ENVI-met	4 days weather data	Assesses indoor and outdoor thermal performance, comparing green walls with bare walls in humid climates.	[136]	
XGBoost, LGBM, RF		ENVI-met	CFD and Building energy data	Cooler microclimates reduce the cooling load of the modeled building.	[133]	
LOR	RNG k- ϵ model	ANSYS Fluent	CFD simulation result	Effectively predicts resident life expectancy and roadside PM _{2.5} concentrations.	[119]	
MLP: Back Propagation Neural Network	Standard k- ϵ model	PHOENICS	1296 CFD data	The integrated framework calculates exposure risks to traffic-related pollutants and plans low-risk commuting routes.	[137]	
XGBoost, RF, EBM	RANS	ENVI-met	72 CFD cases	Analyzes the reduction of particulate pollution near road vegetation.	[36]	
MLP	Modified standard k- ϵ model	PHOENICS	Point cloud data	Develops an automated framework for simulating urban wind environments.	[123]	
MLP		ENVI-met	Pedestrian and weather data	Predicts the impact of street trees on pedestrian comfort and walking behavior	[138]	
RF, XGBoost, DT	Standard k- ϵ model	ANSYS Fluent	680,280 CFD database	Regression models developed to predict urban ventilation efficiency (including green space and water bodies)	[139]	
Surrogate CFD-ML						
ML algorithm	CFD model	CFD tools	Training and test data	Summary/ Key findings	Refs.	
MLP: Back Propagation Neural Network	Standard k- ϵ model, Realizable k- ϵ turbulence model, RNG k- ϵ model	ANSYS fluent	60 cases	Demonstrates high accuracy in predicting wind speed and NO concentrations	[140]	
RF, XGBoost, LGBM	RNG k- ϵ model	Grasshopper AND OpenFOAM	1000 CFD cases	Speeds up traffic pollutant concentration analysis by 800 times in street canyons	[127]	
KNN	k- ω SST	OpenFOAM	80,000 samples	Predicts complex wind flow patterns emerging from street canyons	[115]	
SVM, RT, LR, GPR, CD, ANN, Ensemble	k- ω SST	Grasshopper AND OpenFOAM	300 cases	Rapidly estimates outdoor wind and thermal comfort in large-scale worker villages.	[141]	
ANN: Deep Neural Networks	RANS	OpenFOAM	31 CFD cases	Reduces CFD simulation times by 30% in urban-scale case studies.	[142]	
DT: Extremely Randomized Trees	Modified RNG k- ϵ model		300 CFD cases	Predicts low-speed zones around buildings quickly	[132]	
CNN	Standard k- ϵ model	STAR-CCM+	3400 Wind velocity distribution image	This indicates that building layout is more effective than configuration in mitigating strong winds	[143]	

(continued on next page)

Table 5 (continued)

Surrogate CFD-ML					
ML algorithm	CFD model	CFD tools	Training and test data	Summary/ Key findings	Refs.
GPR	LES	Fluidity	CFD cases	around high-rises. Reduced-order models are six orders of magnitude faster in analyzing urban turbulence.	[144]
GPR	Realizable k-ε turbulence model	ANSYS fluent	159 CFD cases	Predicts wind speeds 10 ⁷ times faster than traditional CFD simulations.	[145]
ANN	RANS	OpenFOAM	CFD database	Cuts costs of designing urban water purifiers by 5.6 to 83.5 times	[146]
RF, DT, ANN, SR	BasinFoam	OpenFOAM	80,000 CFD cases	Provides alternative equations for watershed geometry design.	[147]
LR, SVM, RF, XGBoost	Realizable k-ε turbulence model		83 CFD cases	Predicts pollutant concentrations downwind of vegetation barriers.	[34]
RF		ENVI-met	500 CFD cases	Builds reliable predictive models for outdoor comfort data generation.	[148]
MLP	k-ε turbulence model	OpenFOAM	163 unique geometries	Surrogate model for pedestrian level wind assessment in urban environments	[149]

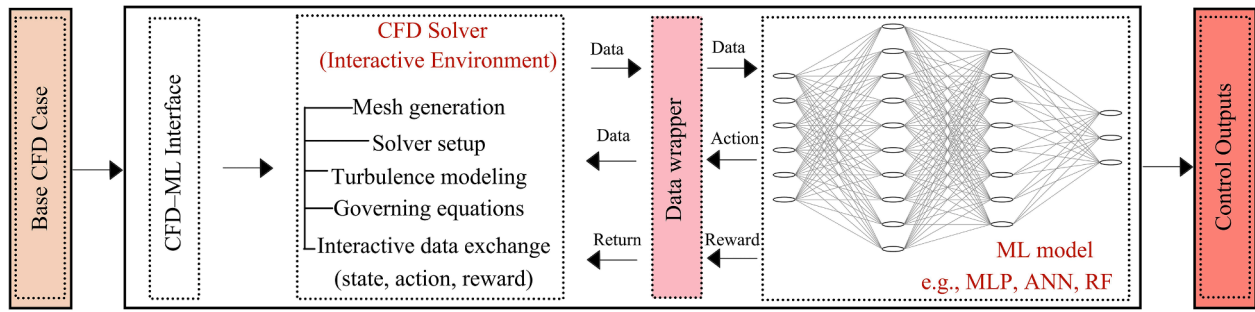
under varying thermal stratifications demonstrated that Convolutional Autoencoder (CAE)-based models combined with LSTM networks achieved prediction errors within acceptable engineering standards, reducing reconstruction time from hours to seconds relative to traditional CFD methods [159]. Despite these advantages, autoencoders require significant computational resources and extensive data for initial training compared to other dimensionality reduction methods [160, 161]. t-Distributed Stochastic Neighbor Embedding (t-SNE), primarily a visualization-oriented nonlinear dimensionality reduction technique, excels at exploratory data analysis and pattern recognition by preserving local similarities in high-dimensional data within a low-dimensional space (typically 2D) [162]. However, due to its non-invertibility, t-SNE is unsuitable for predictive modeling or creating ROMs [163]. Nonetheless, t-SNE effectively reveals hidden structures and similarities within complex datasets, providing valuable insights not immediately apparent from raw data tables [164].

In summary, each method presents unique strengths: POD offers interpretability, autoencoders enhance accuracy, kernel PCA provides precise nonlinear mapping, and t-SNE excels in visualization. A balanced integration of linear and nonlinear dimensionality reduction techniques can effectively optimize computational efficiency, predictive accuracy,

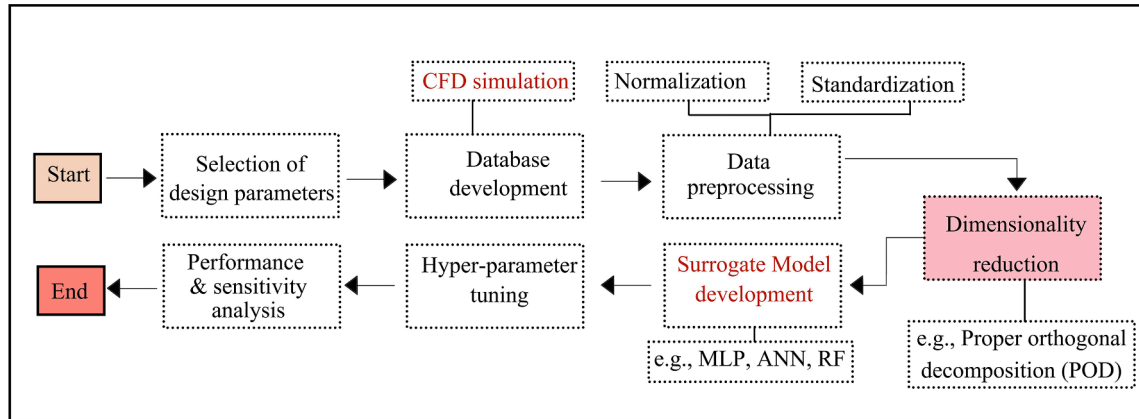
and interpretability, providing robust guidance for future CFD-ML integration in UGI applications.

4.4. Optimizing CFD simulation control using ML

Effective control of UGI is essential for maximizing its benefits in heat mitigation and air quality improvement, and CFD-ML provides powerful tools for optimizing these systems. Among various ML approaches, DRL is a promising approach, enabling adaptive and autonomous decision-making in complex flow environments. By allowing an agent to interact with the environment and learn optimal strategies through trial and error, reinforcement learning is particularly suited to scenarios such as flow control and boundary condition optimization [165]. In CFD simulations, a DRL agent can dynamically adjust boundary conditions, inputs, or shape parameters to optimize specific performance indicators—such as drag reduction, lift enhancement, or improved mixing efficiency [166]. For UGI simulations, this method can help discover configurations that maximize UGI benefits, such as enhanced cooling through evapotranspiration or improved pollutant dispersion via strategic airflow. For example, an RL agent could adaptively control the activation of misting system for green walls or water



(a) Coupled CFD-ML (Bidirectional data exchange)



(b) Surrogate CFD-ML (Unidirectional workflow)

Fig. 7. Generic frameworks for (a) coupled CFD-ML and (b) surrogate CFD-ML.

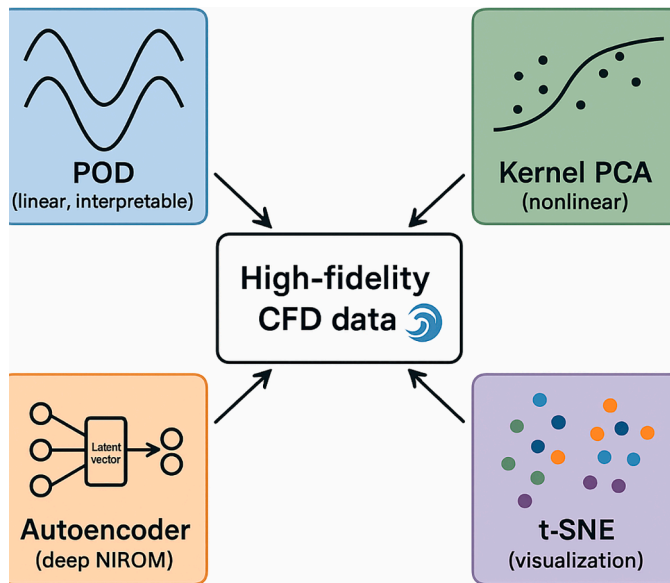


Fig. 8. Data-driven and ML-based dimensionality-reduction techniques for high-fidelity CFD.

features, based on real-time temperature data to reduce urban heat. Unlike traditional control or optimization methods, DRL learns effective strategies directly from high-fidelity CFD data without relying on manual control rules, enabling it to manage complex, nonlinear flow phenomena typical of urban environments.

As one demonstration of DRL's capabilities, researchers have used an artificial neural network to control mass flow rates around a cylinder, achieving about 8% drag reduction [167]. Although this example focuses on a canonical geometry, the methodology can translate to three-dimensional, vegetation-influenced flows, where transfer learning accelerates model convergence and reduces training time by over 50% [168]. Similarly, in turbulent channel flows, DRL-based real-time regulation of near-wall blowing and suction (using microjets) resulted in a friction drag reduction of approximately 37% [169]. These show DRL's potential for controlling complex flows, especially relevant for UGI layouts seeking to optimize pollutant dispersion, wind comfort, and local cooling effects.

Beyond flow control, DRL has been applied to shape optimization [170] and simulation automation [171], highlighting its flexibility in learning control policies directly from simulation data. In an urban context, this data-driven approach could be extended to adaptively manage vegetation arrangement, canopy density, or water feature placement in response to local environmental conditions, thereby maximizing cooling and air quality benefits of UGI. This expands the possibilities for sustainable urban design with minimal human oversight.

4.5. ML in adaptive mesh refinement for enhanced accuracy

Adaptive mesh refinement (AMR) dynamically refines the grid in regions with high flow-field gradients or significant errors, thereby improving simulation accuracy under limited computational resources. This can help in capturing complex flow features around vegetative canopies, vertical greenery, or water surfaces, where steep gradients in velocity, temperature, or pollutant concentrations often occur. However, traditional AMR methods commonly depend on heuristic indicators

(e.g., gradient thresholds) or costly error estimations that require multiple iterative computations, resulting in increased simulation overhead [172,173]. Integrating ML into AMR is expected to yield an intelligent, low-cost strategy for guiding grid refinement, thus balancing accuracy and computational efficiency.

A straightforward approach uses supervised learning to approximate conventional error-based criteria. Given existing flow-field data, neural networks can be trained to directly predict regions requiring refinement, thereby avoiding expensive error evaluations with each refinement step [174]. For instance, Phommachanh [172] employed a multilayer perceptron to learn patterns of complex error-indicator functions; through simple matrix operations, the model can quickly identify where refinement is needed. The results showed that this ML-based approach accurately reproduced the mesh-refinement distributions of high-fidelity error indicators at substantially reduced computational cost, simplifying cumbersome error calculations into a single forward pass. This can be valuable in urban studies where numerous design iterations, vegetation arrangement, species mix, or water feature placement must be tested.

More advanced strategies treat AMR as a reinforcement learning or multi-agent problem. Freymuth et al. [173] proposed an “adaptive swarm mesh refinement” framework, where each mesh cell is treated as an agent, and information is exchanged within local neighborhoods using graph neural networks. By employing multi-agent reinforcement learning, each cell learns whether to refine or coarsen. The results indicated that accuracy remained on par with conventional error-driven AMR while achieving an approximate two orders of magnitude speedup over uniform global refinement.

By reducing the computational cost of targeting critical flow zones, e.g., pollutant plumes swirling around dense foliage or heat fluxes near water bodies, ML-assisted AMR can enhance both the fidelity and scalability of CFD models. These approaches are likely to spur the broader adoption of detailed UGI simulations.

It is critical to emphasize that numerical error evaluation and control are fundamental to ensuring the reliability of CFD results when employing AMR. Without proper error control, AMR may inadequately refine the mesh or incorrectly target mesh refinement regions, leading to non-convergent and potentially unreliable predictions [175]. Consequently, threshold-based refinement criteria have become prevalent. For example, vortex-identification thresholds, such as the Q-criterion, are often used to trigger local mesh refinement around buildings, effectively capturing critical flow structures [176]. Selecting appropriate thresholds necessitates a careful trade-off between accuracy and computational cost: thresholds that are too low improve accuracy but greatly increase computational demand, whereas overly high thresholds can result in significant inaccuracies by missing essential flow details [177].

Moreover, mesh independence analysis and error quantification metrics have emerged as standard practices for quality assurance in urban CFD simulations. As outlined in Section 3.1, GCI is commonly employed to evaluate discretization errors across successive mesh refinements, thus confirming solution convergence with respect to mesh resolution. Recent urban airflow and microclimate studies frequently report GCI values, typically requiring GCI values below 5% to confirm that further mesh refinement yields negligible improvement [178]. This involves comparing results from two or more mesh resolutions; convergence is assumed when the change in critical outcomes between refined meshes falls beneath a predefined threshold (e.g., 5%). Additionally, iterative convergence criteria are applied to control solver residuals, ensuring each adaptive mesh refinement step achieves fully converged solutions. Residual thresholds of 10^{-5} to 10^{-6} for continuity and momentum equations have been recommended and adopted in recent CFD studies focusing on urban environmental simulations, thereby ensuring numerical reliability [179,180].

Integrating these quantitative error criteria including threshold-based refinement, GCI-based mesh independence, and residual convergence metrics significantly enhances the credibility and quality of urban microclimate and UGI-related CFD simulations.

4.6. Other emerging technologies at the intersection of CFD and ML

The integration of CFD and ML has also advanced in several emerging directions, expanding the theoretical foundation of Coupled CFD-ML and Surrogate CFD-ML. These include Physics-Informed ML, turbulence modeling augmented by ML, and flow-field super-resolution reconstruction.

In Physics-Informed ML, the governing equations of fluid mechanics are directly embedded into ML models to enhance physical consistency and generalizability. A representative approach is the Physics-Informed Neural Network (PINN), which incorporates residuals of the Navier–Stokes or other governing equations into the training loss function, ensuring that the learned model satisfies physical laws while fitting the data [181]. Compared to purely data-driven models, physics-guided approaches inherently enforce conservation principles (e.g., mass and momentum conservation), exhibiting greater robustness under unseen conditions [182]. For example, some studies have used PINNs with limited wind-tunnel velocity data to successfully reconstruct airflow patterns around buildings and estimate the associated surface pressure distributions [183]. This integrated method can “mine” hidden flow-field information from sparse data. In the context of UGI, such approaches enable robust yet data-efficient modeling of airflow and pollutant transport around vegetation and water features.

Another prominent research area is improving closure accuracy in turbulence simulations, such as RANS or LES, by incorporating ML. The field inversion and ML methodology employs high-fidelity (or experimental) data to invert optimal correction fields in turbulence models, and an ML model is then trained to capture the relationship between these corrections and local flow variables [184]. Halder [185] introduced a data-driven turbulence model that augments a traditional RANS model with a correction term learned via ML, thereby offering better predictions of experimentally observed complex flow phenomena. This approach embeds a trained ML submodel into the CFD solver, providing real-time corrections for turbulent viscosity or Reynolds stress. Such an approach significantly improves predictions in separated, vortical, and complex flow regimes. For UGI applications, such turbulence modeling enhancements can more accurately capture canopy-induced turbulence and pollutant dispersion, guiding effective vegetation placement.

Beyond these techniques, other novel methods exist for coupling CFD and ML. For instance, deep learning can facilitate flow-field super-resolution reconstruction by inferring high-resolution solutions from coarse-grid simulations, utilizing low-resolution CFD results to generate fine details [186]. Meanwhile, neural operators approximate solutions to fluid mechanics’ governing partial differential equations, enabling rapid flow field predictions under various initial and boundary conditions [187]. These methods can be viewed as new-generation CFD surrogate models that integrate physical priors with high-dimensional function-approximation capabilities, representing a compelling extension of traditional surrogate modeling. Applied to UGI design, such surrogate models can rapidly evaluate different greening configurations.

5. Application areas of CFD and ML integration in UGI research

Integrating CFD and ML enhances urban environmental assessment by employing their complementary strengths. This combination ensures high-precision ML predictions while accounting for critical factors such as urban geography, surrounding buildings, vegetation parameters, UGI types, and climatic conditions. Consequently, simulations become more comprehensive, efficient, and capable of addressing complex urban scenarios.

Fig. 9 shows four areas where CFD-ML integration can be effectively applied in urban environments: evaluating the best design, accelerating simulation, improving physical understanding, and enhancing the quality of CFD simulation. For instance, ML enhances CFD simulation accuracy through automated image processing [114,188], predicts high-pollution zones, and identifies pedestrian paths with optimal

comfort based on CFD data [137]. ML also forecasts UGI impacts on pedestrian walking patterns [138]. Algorithms like RF and SVM analyze vegetation characteristics, including dust retention and spectral properties [189]. This integration accelerates simulations, optimizes UGI design, and improves physical understanding by streamlining processes and refining environmental predictions. This section explores different CFD-ML tool combinations for five UGI types.

5.1. Street trees (ST)

ST are essential components of UGI for air quality improvement and heat mitigation. Integrating CFD and ML enhances modeling and prediction of street trees' environmental benefits. Recent studies have applied CFD-ML techniques to optimize the pollution reduction and microclimate effects of roadside vegetation.

In air quality research, surrogate CFD methods combined with ML have been widely applied to evaluate the effectiveness of street vegetation in pollution mitigation. For example, Hashad et al. [34] utilized a CFD-ML integration to assess how roadside vegetation barriers influence pollutant dispersion and deposition. Their approach began by constructing an ML training dataset derived from 83 RANS CFD simulations, which systematically varied vegetation parameters, including LAD, along with computational domain and boundary conditions to realistically model aerodynamic and sedimentation processes. The subsequent ML model incorporated key features such as vegetation width, height, LAI, particle size, and local wind speed (Fig. 10). Model evaluation indicated excellent predictive performance, with neural networks, RF, and XGB algorithms achieving an average R^2 of ≥ 0.91 within computationally rapid times (approximately one second).

Similarly, other studies applied CFD-ML techniques to street tree scenarios involving pollutant transport from roadways to residential

zones. For instance, logistic regression models trained on CFD simulation results from ANSYS Fluent accurately estimated $PM_{2.5}$ concentrations with less than 10% error compared to measured data [119]. Another noteworthy integration utilized the microclimate model ENVI-met coupled with ML algorithms, providing enhanced multifactorial analyses of urban air quality [36]. The approach involved creating structured input matrices from ENVI-met simulations for subsequent ML analysis (Fig. 11). Through this integrated method, researchers quantitatively examined interactions among urban structures, street vegetation patterns, area height, and wind direction, collectively explaining approximately 16.1% of particulate concentration variance. Again, RF and XGBoost algorithms demonstrated strong predictive accuracy, achieving R^2 values consistently exceeding 0.90, in alignment with findings from Hashad et al. [34].

Beyond air quality modeling, CFD-ML approaches have proven beneficial in assessing thermal mitigation effects of street trees, particularly regarding wind flow modifications and air temperature reductions. Effective CFD modeling of urban vegetation typically requires detailed semantic and geometric data, often derived from remote sensing techniques. Integrating ML, specifically through advanced semantic segmentation methods, significantly enhances the precision and computational efficiency of these models. Sun et al. [123] exemplified this integration by developing an automated deep learning framework based on urban point cloud data to efficiently generate CFD-ready models of complex urban environments, including street-level vegetation (Fig. 12). The framework encompasses four stages: data collection with RGB point cloud generation, semantic classification using the Pointnet++ deep learning model, geometry simplification via GPR and clustering algorithms, and rapid CFD simulations with Phoenix software. Results demonstrated that this streamlined process accurately and efficiently predicts changes in urban wind speeds and air temperatures,

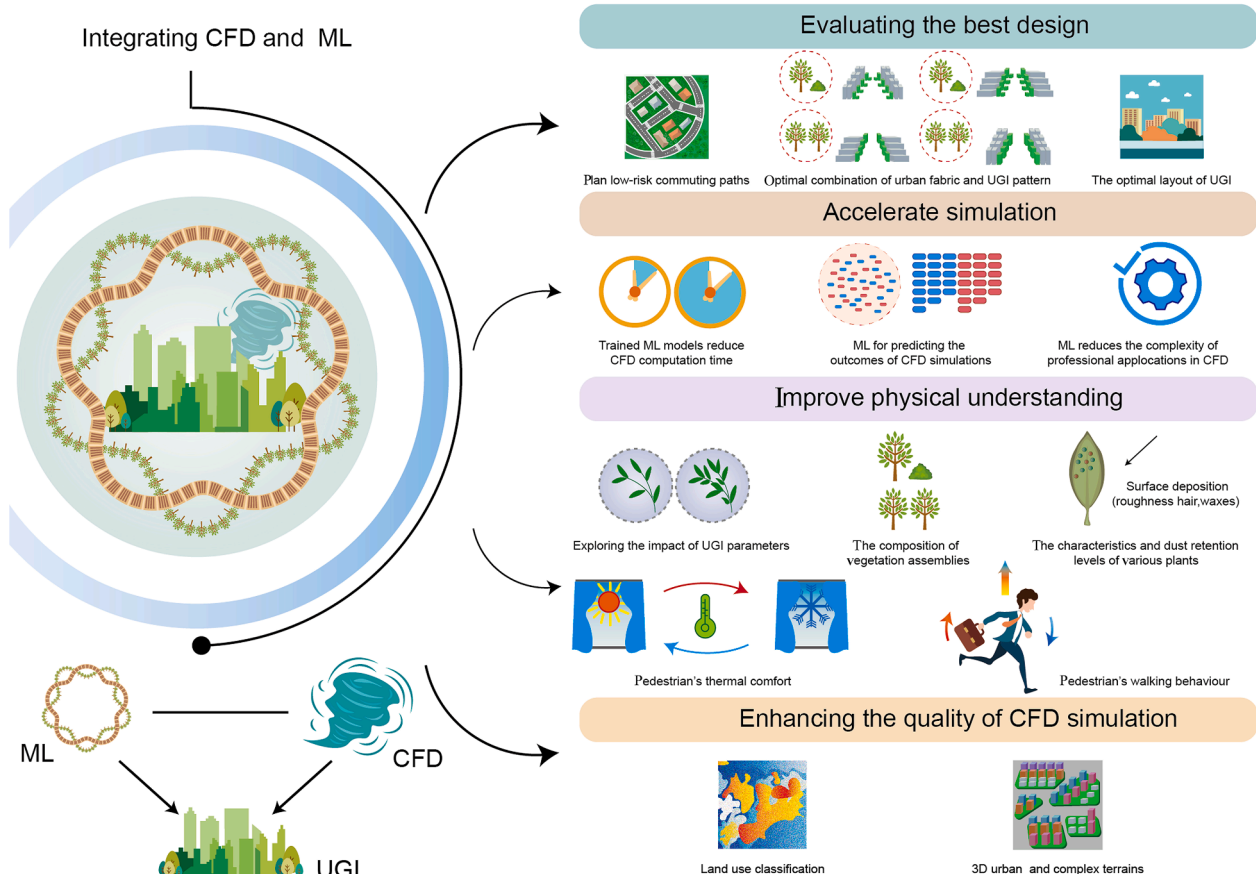


Fig. 9. Application areas for integrating CFD and ML in UGI.

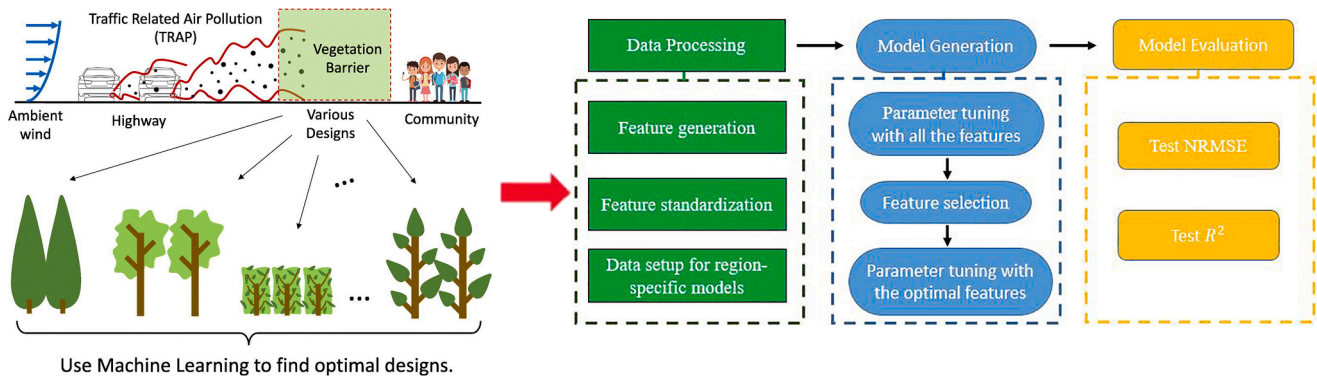


Fig. 10. Utilizing CFD-ML to design roadside UGI that reduces traffic-induced air pollution adapted from [34].

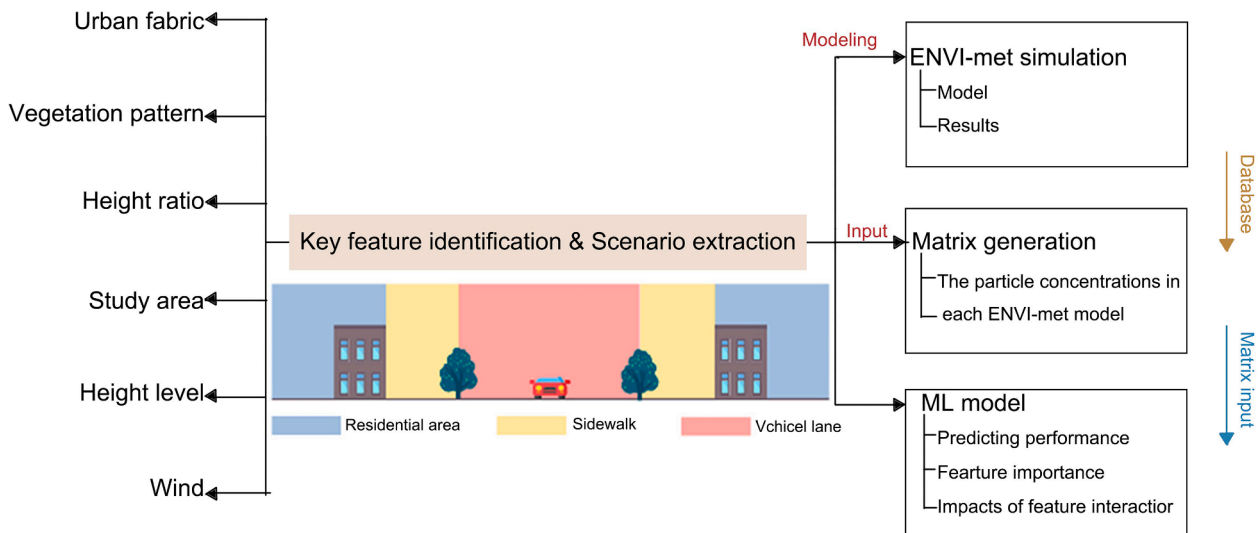


Fig. 11. Coupling ENVI-met and ML to analyze the impact of urban vegetation configuration on particulate matter concentration, adapted from [36].

particularly beneficial during heat wave conditions when timely interventions are critical.

5.2. Urban forest and parks (URP)

Building on the application of CFD and ML in street tree analysis, URP offers another significant avenue for improving urban cooling and air quality. However, studies fully integrating CFD and ML to analyze URPs remain limited. Specifically, the inherent complexity arising from high plant species diversity and heterogeneous three-dimensional canopy structures in URPs significantly complicates CFD modeling, necessitating exceptionally high-resolution grids and advanced turbulence models. Furthermore, the substantially larger spatial scales of urban parks compared to street-level or building-scale green infrastructures further escalate computational requirements, thus restricting the widespread integration of CFD and ML methodologies in URP studies. This section reviews existing studies that have primarily focused on optimizing cooling loads within adjacent built environments and evaluating vegetation dust retention capabilities.

Alyakoob et al. [133] demonstrated a sophisticated application of CFD and ML integration in analyzing microclimatic conditions and associated cooling demands around three university buildings in Arizona. Utilizing the ENVI-met model, they developed detailed microclimate simulations incorporating precise building geometries, lawn layouts, and diverse tree distributions (Fig. 13). Subsequently, they applied ML techniques to predict baseline cooling loads under varying urban park microclimates. Among the tested algorithms, RF exhibited

superior predictive performance compared to XGBoost and LGBM, achieving accuracies up to 0.98. RF's superior performance was attributed to its inherent robustness, eliminating the need for parameter scaling during data preprocessing and effectively identifying critical parameters that significantly influenced cooling load predictions. This integrated approach illustrates the substantial impact of urban forest and park configurations on adjacent building energy efficiency and highlights the importance of precise microclimate modeling in energy-saving strategies.

Expanding on the application of ML-driven efficiency in thermal analysis, Ahn et al. [190] introduced an advanced CFD-ML methodology targeting large-scale thermal modeling of urban parks. By combining Rhino 7.0 and STAR-CCM+ software with the Design Manager system of STAR-CCM+, their approach incorporated a data augmentation model capable of rapidly generating extensive training datasets. Starting from an initial dataset of only 100 randomly generated data points, the augmented dataset expanded to hundreds of thousands of simulations required for robust ML training. This significantly reduced computational demands, enabling urban thermal environment analyses to be completed within roughly one hour—remarkably faster than conventional approaches, which typically required more than 400,000 h. Such computational efficiency represents a crucial advancement, allowing timely optimization of vegetation configurations within urban parks for enhanced thermal comfort and cooling efficiency.

Transitioning to air quality considerations, Zhang and Zhang [189] presented an innovative ML framework focused specifically on quantifying and optimizing vegetation dust retention capabilities within urban

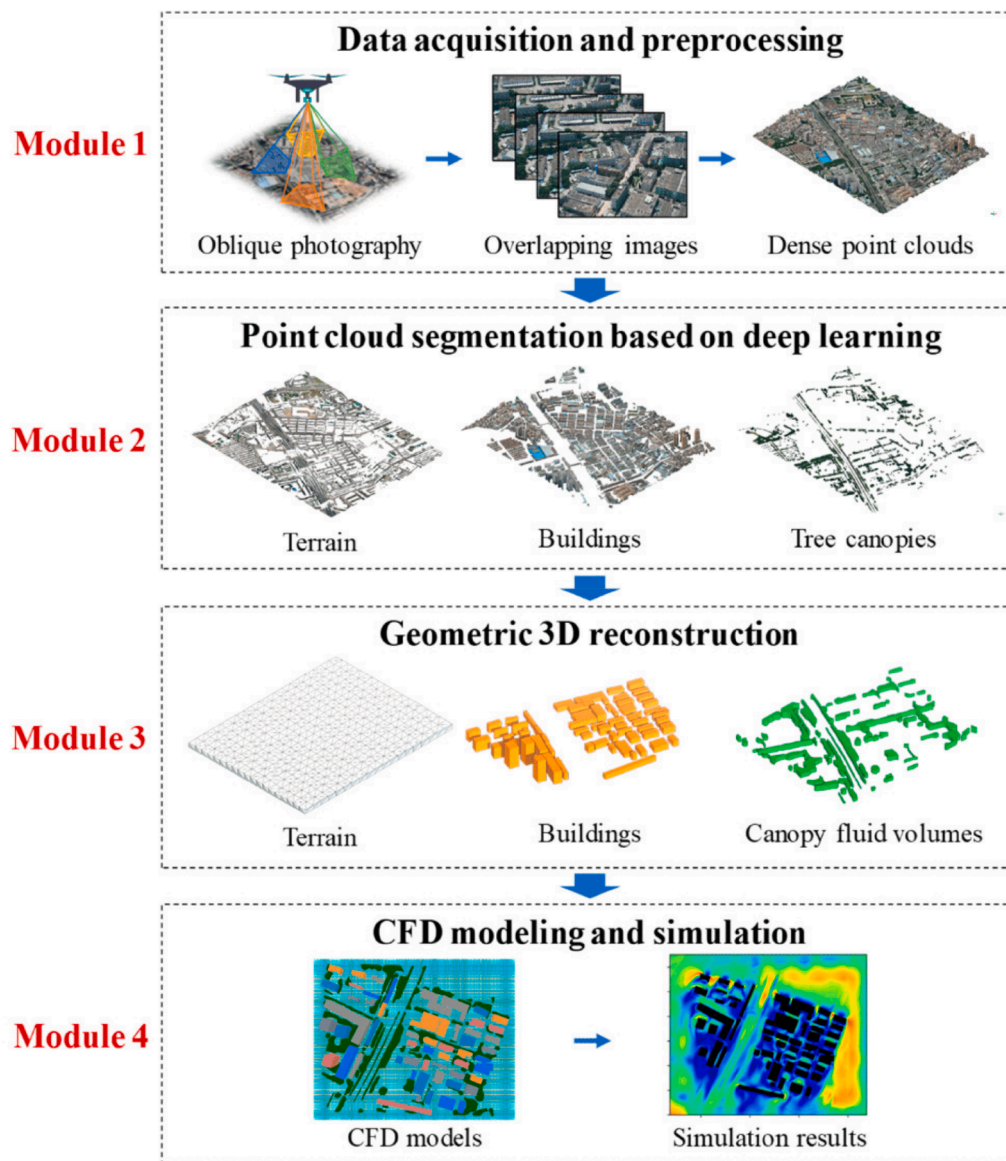


Fig. 12. Integrated CFD and ML image segmentation methods [123].

parks. Utilizing spectral reflectance data and empirical dust retention measurements, their study applied RF algorithms to accurately predict dust retention capacities across different plant species, complemented by SVM algorithms for classifying vegetation into high and low dust-retention groups. This ML-based methodology significantly outperformed traditional correlation-based approaches due to its capacity to handle complex, nonlinear relationships inherent in vegetation data. Their findings confirmed strong correlations between plant spectral characteristics and dust retention efficiencies, highlighting ML's potential to inform optimized vegetation selection strategies for air quality improvement in urban park settings. Further integration of these ML insights with CFD simulations could enable more precise and efficient planning for air quality management, maximizing environmental benefits.

5.3. Green wall (GW) and green roof (GR)

GW and GR are key components of UGI, known for enhancing energy efficiency, thermal comfort, and air quality. The integration of CFD and ML offers a promising approach to optimize their performance. This section reviews studies employing ML and simulation techniques to

improve the design and efficiency of green walls and roofs, emphasizing cooling potential, energy savings, and pollutant removal.

Daemei et al. [136] investigated the thermal performance of green walls by integrating deep learning algorithms with EnergyPlus and ENVI-met simulations. Their study utilized a Multilayer Perceptron (ANN), trained on experimental data, to rapidly predict GW performance compared to bare wall conditions. Specifically, EnergyPlus was applied alongside a GW-specific module to simulate indoor thermal conditions, while ENVI-met provided outdoor microclimatic simulations. The ANN achieved a high prediction accuracy, estimating air temperature and relative humidity with only 0.6% error relative to simulation outcomes. Results demonstrated that the green wall significantly improved thermal conditions indoors, reducing air temperatures by approximately 9% and humidity by around 32%. Outdoors, GW lowered ambient temperature by 0.36 °C and marginally increased relative humidity by 1.04%. This integrative CFD–ML method underscores GW's potential as a viable cooling solution and highlights the effectiveness of machine learning for rapid and reliable performance prediction.

Extending this integrative approach to green roofs, Mousavi et al. [191] proposed an optimization framework specifically designed to

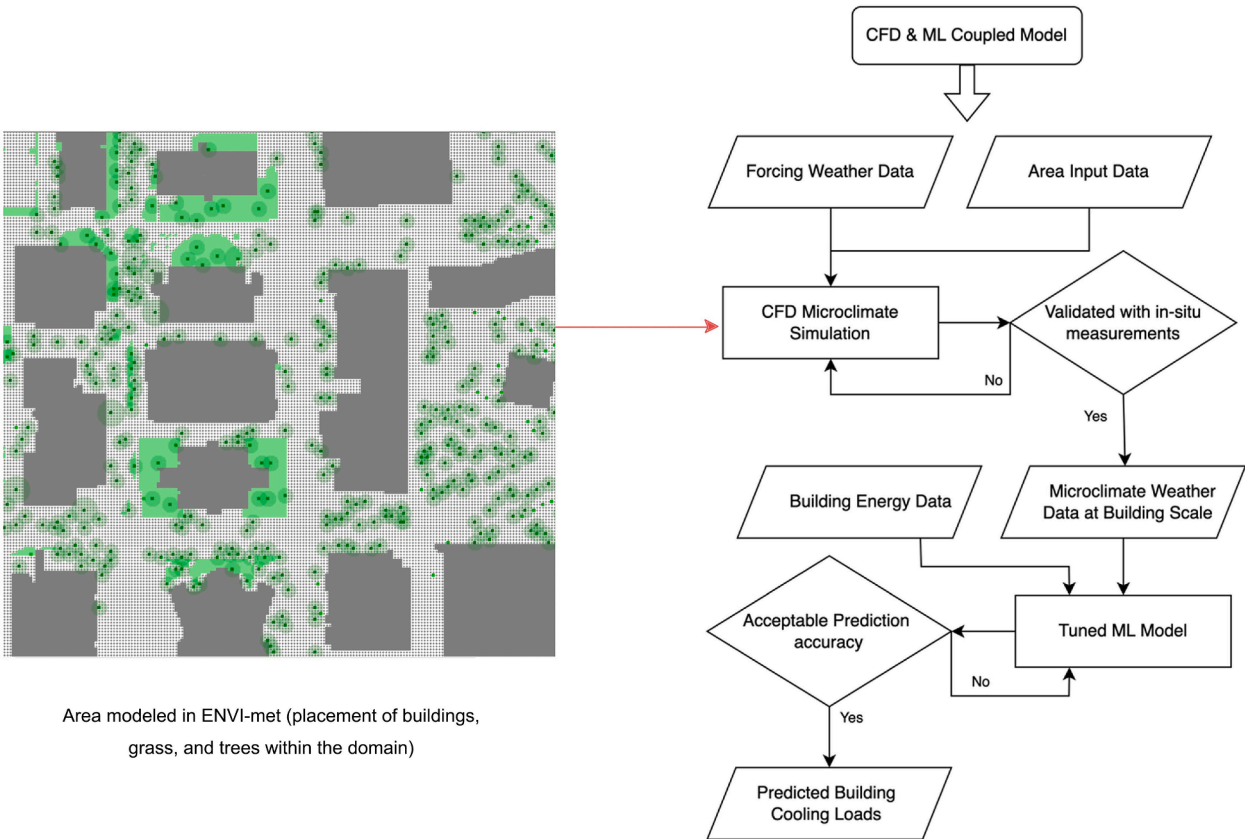


Fig. 13. Integrated CFD and ML predicted microclimate and cooling loads [133].

enhance energy savings and thermal comfort in buildings (Fig. 14). Their model employed four clearly defined objective functions alongside key GR parameters, utilizing an adaptive neuro-fuzzy inference system (ANFIS) hybrid learning algorithm integrated with DesignBuilder simulation software. The optimized model yielded predictions with a

Root Mean Square Error (RMSE) under 0.5, validating its reliability and precision. Leaf emissivity emerged as the most influential parameter for GR design optimization, contributing significantly to both energy savings (a reduction of approximately 14%) and improved occupant comfort (comfort time increased by 12.8%). By effectively combining

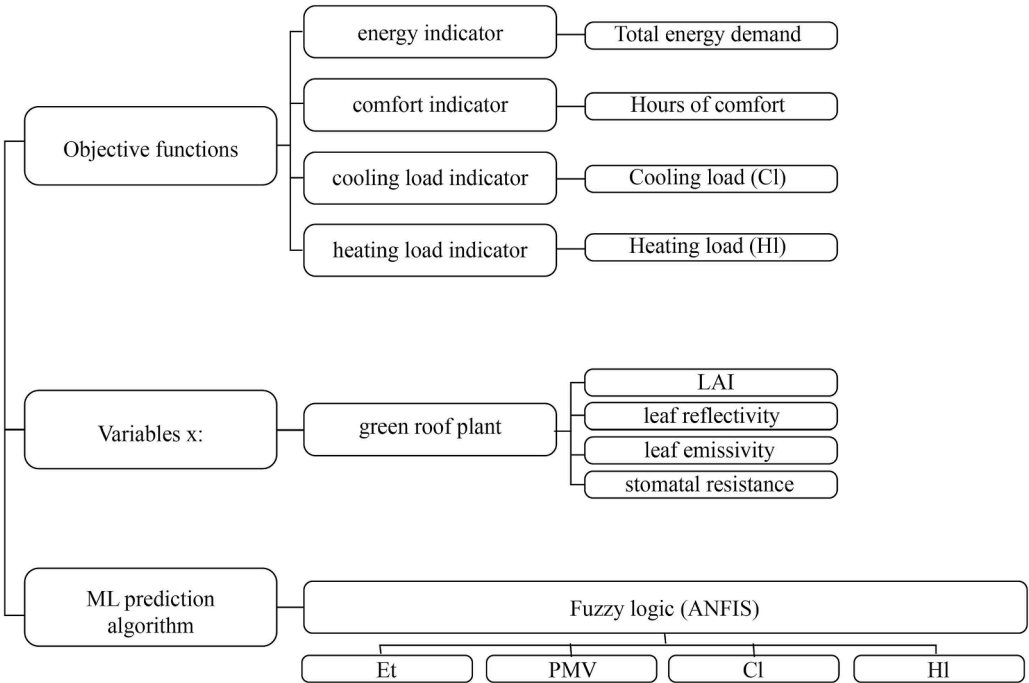


Fig. 14. GR optimization design model for multi-tool collaboration [191].

MATLAB's ANFIS capabilities with numerical simulations, their integrated approach provided a powerful methodology to accurately determine optimal GR designs, achieving significant building-level environmental benefits.

Furthermore, accurately identifying suitable areas for GR implementation is critical for assessing their large-scale air quality improvement potential. Traditional field-based measurements are often impractical or inefficient for expansive urban areas, while conventional remote sensing image classification approaches frequently result in fragmented or imprecise outcomes. Addressing this, Huang [192] developed a spatial distribution model for GR using an advanced deep learning image segmentation technique (U-Net), combined with Python-based simulations. This integrated approach rapidly and accurately identified potential GR implementation areas throughout large urban environments, achieving a validation accuracy exceeding 94% and completing comprehensive city-wide mappings within just 20 min. In contrast, earlier attempts using alternative algorithms such as SVM yielded considerably lower classification accuracy (approximately 80%) [193]. These findings emphasize the importance of integrating precise segmentation methods with CFD modeling to effectively delineate vegetation boundaries from urban structures. Such integration is essential to accurately capture key physical phenomena, such as vegetation-induced subsidence and turbulence effects, in comprehensive air quality analyses.

5.4. Water features and aquatic vegetation, urban water (UW)

Urban water features, including ponds, rivers, and aquatic vegetation, play an essential role within UGI due to their notable impacts on urban microclimates. Integrating CFD with ML has emerged as a

promising strategy to optimize the design and operational performance of these water bodies, particularly by improving simulations of pollutant dispersion, heat exchange, and fluid dynamics. Recent research has demonstrated the effectiveness of CFD–ML frameworks in enhancing accuracy, reducing computational complexity, and supporting informed decision-making in urban water design.

Li and Sansalone [146] developed an efficient and cost-effective CFD–ML optimization methodology specifically aimed at urban clarification basins. Their approach combined CFD simulations conducted via OpenFOAM, ML model development through PyTorch, and optimization procedures using Dakota software (Fig. 15). Initially, detailed CFD simulations using OpenFOAM generated a comprehensive performance dataset of various basin configurations. Subsequently, ANN implemented through PyTorch was trained using this CFD-generated database. Employing Dakota's optimization library, the ANN-driven model significantly refined the basin designs. This integrated CFD–ML approach achieved remarkable predictive accuracy ($R^2 = 0.998$) in evaluating particulate matter clarification efficiency, while simultaneously offering substantial computational and cost savings. Remarkably, the entire ML optimization process could be completed within minutes on standard personal computing hardware, reducing traditional design costs by factors ranging from 4.7 to 8 times.

Further advancements in CFD–ML integration were demonstrated by substituting conventional CFD simulations entirely with ML algorithms. For instance, a composite neural network integrating Physics-Informed Neural Networks and Deep Operator Networks was employed to predict fluid dynamics and particulate transport within clarification basins [194]. This innovative approach directly embeds physical principles into neural network structures, significantly accelerating model training and minimizing GPU memory demands. Despite these computational

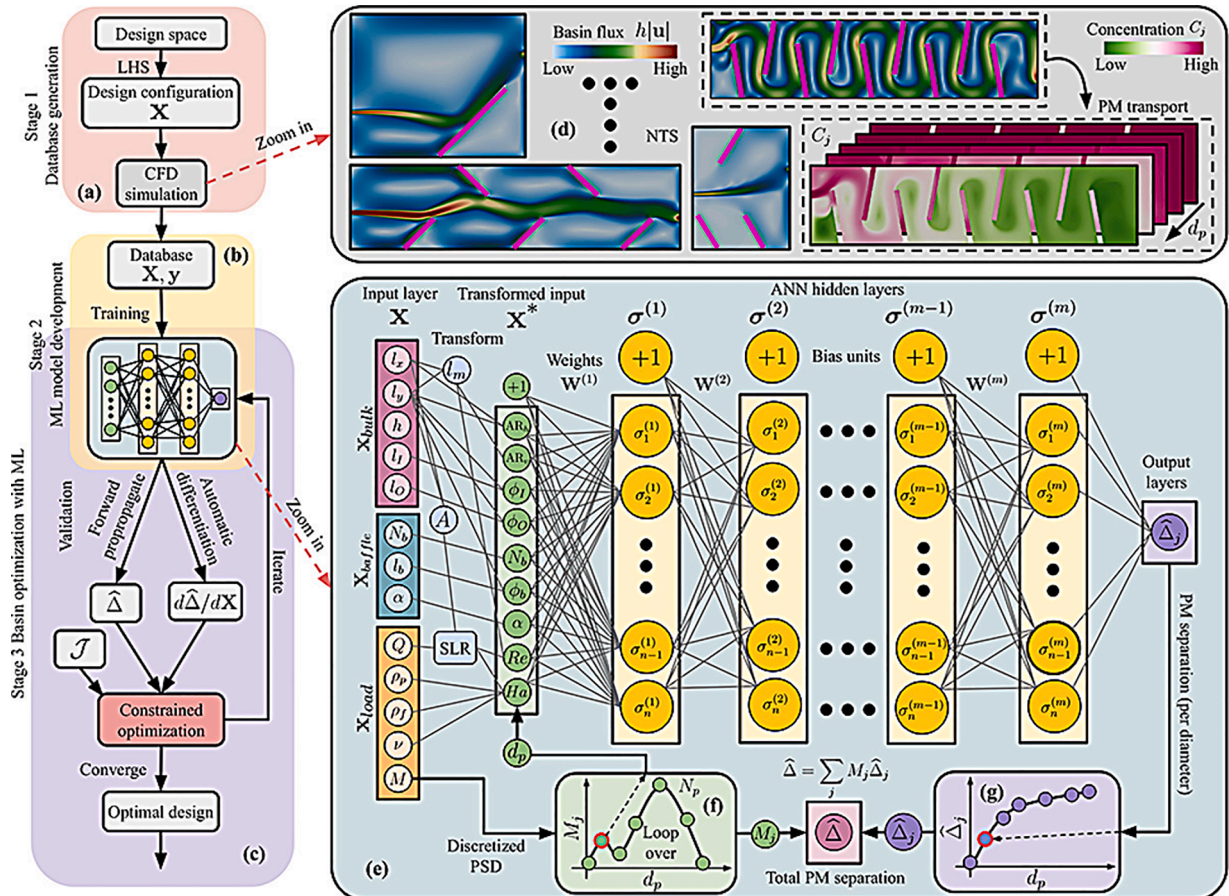


Fig. 15. Developing and applying a data-driven ML model framework for basin performance inference and optimization [146].

advantages, the method required extensive training data (approximately 80,000 training cases), and predictive accuracy exhibited variability, with only 66.4% of predictions achieving R^2 values greater than 0.8. While not explicitly evaluating impacts on urban thermal environments or air quality, this approach holds potential value by optimizing geometric configurations of urban water features. Such geometric optimizations can indirectly enhance microclimatic conditions in adjacent urban areas by influencing local heat exchange processes and airflow patterns.

6. Discussion

6.1. Methods and tools for CFD-ML integration in UGI applications

The literature review reveals that integrating CFD and ML has enabled researchers to develop efficient, dynamic predictive models, effectively capturing UGI impacts and significantly reducing computational costs. This integrated framework can be highly automated, facilitating advanced spatial prediction systems with considerable temporal and resource efficiency. However, this integrated approach still faces a shortage of extensive CFD simulation data necessary for training ML models, particularly as high-precision, urban-scale CFD simulations demand more advanced equipment and time. The integration of CFD and ML in UGI applications is still developing, necessitating choosing between Coupled CFD-ML and Surrogate CFD-ML. Notably, studies directly substituting CFD with ML demonstrate lower accuracy than those employing high-fidelity CFD simulations. Hence, ML should complement rather than replace CFD simulations. Supporting techniques such as dimensionality reduction, DRL, AMR, physics-informed machine learning, and flow-field super-resolution reconstruction can serve as auxiliary methods to significantly enhance CFD efficiency, accuracy, and simulation control.

Regarding CFD simulations for ML training, CFD effectively models complex interactions in urban microenvironments. However, due to diverse mechanisms and parameters from different UGI types, the impacts of varying UGI parameters still require investigation. Vegetation modeling complexity further increases CFD simulation specificity. For instance, ST and URP feature complex, heterogeneous canopies, increasing geometric complexity and environmental variability. Their effects on thermal mitigation and air quality depend heavily on interaction mechanisms, especially given the strong influence of street geometry and ventilation conditions [78,93]. Seasonal changes compound these variabilities, potentially reducing ML model accuracy without timely updates. Based on findings in Sections 5.1 and 5.2, CFD-ML integration has successfully predicted short-term localized spatial scenarios. However, in short, tree-dominated UGI (ST, URP) is feasible for ML-CFD integration at small scales [34,36,119], but it faces greater complexity in capturing 3D canopy effects and ensuring model robustness across different sites and conditions.

In contrast, GR and GW offer more controlled environments with simpler, more uniform geometries. Roof or wall vegetation typically forms planar structures with well-defined substrate and vegetation layer characteristics, simplifying airflow obstruction and turbulence compared to individual trees [79–81]. This relative simplicity makes GR/GW highly suitable for CFD-ML approaches at the building scale. Data collection for GR/GW typically relies on building-monitoring systems (e.g., temperature, energy consumption) or targeted experiments, providing relatively clean datasets for ML training [136,144]. However, extending GR/GW solutions beyond single buildings introduces new complexities, particularly regarding whether city-scale airflow interactions (e.g., multiple green roofs altering wind directions) can be accurately represented solely by ML without oversimplification.

UW, such as ponds, lakes, and vegetated waterways, represent a unique UGI category characterized by multi-physics complexities. While geometrically simpler than forested areas, accurately modeling UW impacts necessitates coupling hydrodynamics of water and air with heat

and moisture exchanges, occasionally incorporating water quality and aquatic vegetation effects. Consequently, high-fidelity CFD simulations for UW are computationally demanding. Here, integrating ML significantly enhances predictive speed, enabling nearly instantaneous outputs for optimization purposes [146]. However, studies using deep neural networks to replicate CFD-based flow and pollutant transport required around 80,000 training scenarios, yet only approximately two-thirds achieved acceptable accuracy ($R^2 > 0.8$) [194]. Despite substantial datasets, performance remains suboptimal, highlighting increased data density and model complexity requirements compared to air-only scenarios. Additionally, hydrological studies remain scarce, likely due to specialization and difficulties in obtaining comprehensive training data (limited urban waterscape measurements and high CFD computational costs). Environmental variability (e.g., temperature, water level, water quality fluctuations) also necessitates frequent model retraining or physics-based architectures to maintain effectiveness under changing boundary conditions.

Currently, ENVI-met is widely chosen due to its robust vegetation library, detailed characteristics, and vegetation densities. For enhanced accuracy, OpenFOAM simulations integrated with Rhino and Grasshopper plugins utilize point cloud data to study vegetation impacts on microclimates. Furthermore, Grasshopper facilitates easy integration with Python and genetic algorithms to establish new workflows, enabling rapid evaluation of the optimal UGI designs. As for parameter refinement, the built-in Design Manager in STAR-CCM+ automates multiple design iterations in batch mode, adjusting simulation parameters (e.g., environmental factors), objectives (e.g., extreme temperature thresholds), and constraints to explore various configurations [190]. However, coupling multiple software packages remains challenging. Advances in data science coupling methods will further improve ML integration in urban microenvironments. Thus, combining professional CFD simulation software with ML-appropriate model software is a potential direction for integrating CFD and ML in UGI research.

6.2. Algorithm selection for CFD-ML integration

The review indicates that the selection of machine learning algorithms largely depends on specific application requirements. When selecting an appropriate algorithm, researchers should refer to the algorithm applications summarized in Fig. 16, evaluate the technical characteristics of each algorithm category (Fig. 6), and consider factors such as dataset size (Table 5), interpretability needs, and specific application scenarios (Fig. 9). From a technical perspective, when analyzing interactions among multiple factors involved in pollutant removal and microclimate regulation by UGI, it is critical to consider how different algorithms handle nonlinear relationships. Each ML algorithm utilizes distinctive mechanisms, such as activation functions, ensemble averaging, kernel mapping, and instance-based reasoning, to manage nonlinearity effectively. This review discusses four main algorithm categories and their representative algorithms as follows:

Neural network-based algorithms: MLP is commonly employed in this category. MLP comprises neurons that apply weighted sums to inputs and pass them through nonlinear activation functions, enabling the learning of relationships far beyond simple linear correlations [35,195]. Within UGI research contexts, such as predicting pollutant dispersion or cooling potential, MLP uses its hidden layers to automatically learn complex combinations of factors, for instance, optimizing cooling in urban areas adjacent to green spaces [125] and analyzing how varying parameters of GR influence thermal comfort [195]. However, despite its strong potential to capture high-order feature interactions, careful attention is necessary to avoid overfitting and to maintain physical interpretability.

Tree-based algorithms: RF, a representative tree-based algorithm, effectively addresses nonlinearity through an ensemble of decision trees. Each tree partitions data according to thresholds defined on input features, and RF aggregates predictions from many such trees via ensemble

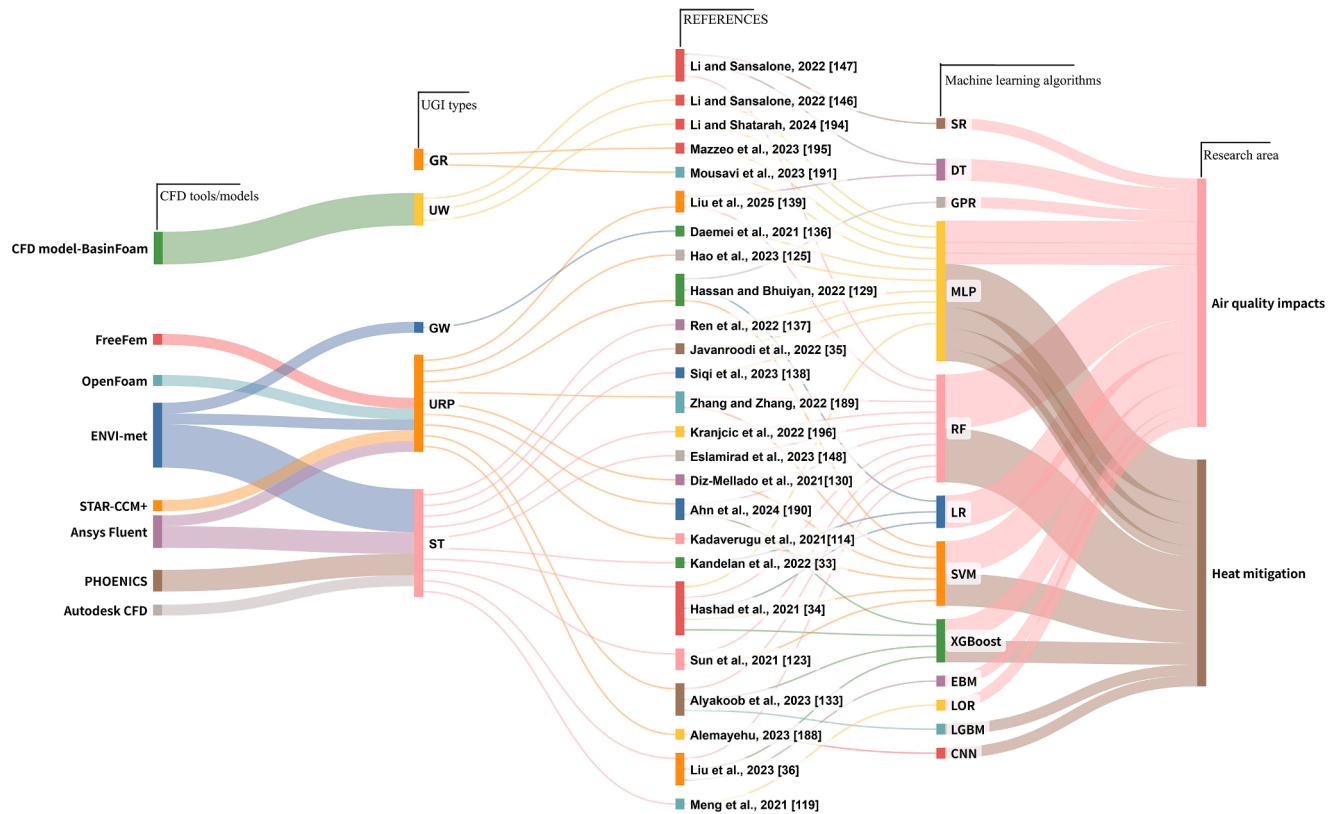


Fig. 16. Sankey diagram showing the reviewed articles on integrating ML and CFD methods and stand-alone ML in UGI research during 2020–2025.

averaging, thus resisting overfitting and simultaneously capturing complex interactions among high-dimensional features [36]. For example, RF has been successfully applied to identify optimal tree planting locations based on urban land cover for air pollution mitigation [196]. In studies examining the impacts of UGI on microclimatic variables and energy consumption, RF robustly analyzes complex nonlinear interactions among multiple independent variables, without requiring stringent data preprocessing [127,133].

Kernel-based algorithms: SVM and GPR handle nonlinear relationships by employing kernel transformations. These algorithms map input features into higher-dimensional spaces, where linear patterns become easier to fit, thus effectively modeling intricate nonlinear relationships inherent in original variables [141]. Previous studies have demonstrated SVM's successful application of kernel methods in integrating complex interactions among building morphology, vegetation, and meteorological parameters [197]. Similarly, GPR utilizes kernel-defined covariance functions to infer smooth, non-parametric functions fitting the data, and provides uncertainty estimates that are beneficial for capturing variability in urban wind patterns [144,145]. However, a notable trade-off is that kernel-based methods scale poorly as dataset size grows, so they can impose significant computational burdens for large number of training samples; dimensionality reduction or sparse/approximate kernels are often needed.

Regression-based algorithms: More traditional regression algorithms serve as valuable baselines and interpretable models, though they often face limitations in modeling nonlinear relationships encountered in UGI research. Among these, KNN is an instance-based learner providing a simple yet powerful approach to local nonlinear modeling. KNN regression does not assume a global functional form; instead, it predicts outcomes by averaging the results of the k most similar instances [115]. This local characteristic implies that KNN can model relationships of arbitrary complexity, provided sufficient data density. However, a primary drawback of KNN is its limited extrapolation capability, as predictions are confined to areas with adequate local observations.

Consequently, in practical applications, regression-based algorithms are commonly employed as benchmark references [119,129] or as preliminary approaches, often complemented by more sophisticated nonlinear models such as RF, MLP, or SVM.

As shown in Fig. 16, most algorithms apply to both air quality impacts and heat mitigation. However, algorithms like DT, SR, GPR, and LR are mainly used for air quality studies, whereas CNN and LGBM are predominantly employed in heat mitigation research. Despite the advancements made, many review articles still lack a clear rationale for the selection of specific ML algorithms and do not provide a systematic comparison of different approaches. To encourage broader adoption of ML in urban studies, it is vital to enhance interpretation and understanding, enabling researchers to effectively select suitable algorithms for complex urban challenges.

Additionally, ensemble learning methods warrant particular attention due to their potential in improving prediction accuracy. These methods can generally be categorized into homogeneous (internal) and heterogeneous (meta-model-based) ensemble approaches [198]. Homogeneous methods typically combine multiple instances of the same base algorithm to form a unified predictor, aiming to reduce variance or bias. Representative examples include RF and XGBoost, which employ bagging multiple decision trees or sequentially re-weighting weak learners (boosting), thus significantly enhancing accuracy [133,199]. In contrast, heterogeneous ensemble methods, such as stacking and weighted voting, integrate predictions from different algorithms using an external meta-learner or optimized weighting schemes [200]. This approach often yields predictions that are more stable and generalizable across various conditions. For instance, the "Stacking-EML" framework, integrating five distinct ML algorithms with an additional meta-learner, achieved notably higher predictive skill (R^2 up to ~ 0.99) in regional climate predictions compared to single-model or traditional climate model outputs [201].

However, these advantages come at a cost. Increased model complexity can diminish interpretability—for example, aggregating

numerous decision trees might obscure clear insights into feature importance. Moreover, training and tuning multiple learners, as well as an additional meta-learner, can be resource-intensive, particularly because stacking typically trains each base model K times across K -folds to generate out-of-fold predictions (and often retrains on the full dataset for the final model) [202]. This complexity likely contributes to the limited adoption of meta-model ensemble methods within current UGI-focused CFD research. In summary, while combining multiple models can significantly boost performance, explicit application of ensemble methods to enhance prediction accuracy and reliability specifically in urban environmental and CFD contexts remains relatively unexplored.

6.3. Limitations and potential solutions

Integrating CFD and ML offers great potential for reducing computational costs and improving the design of UGI based on simulation results. However, there are several limitations and areas for improvement in current integration strategies. One key issue is the requirement for large amounts of training data in ML models, particularly in the Surrogate CFD method. The accuracy of predictions depends heavily on the availability of vast CFD simulation data, which can be costly to generate and often requires high-performance computing resources. Ensuring access to sufficient data is critical to advancing the field.

Due to the high computational cost associated with CFD simulations, researchers frequently encounter data scarcity challenges. To address this issue, several strategies have emerged, including synthetic data generation and data augmentation. Generative models, such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), enable the creation of realistic synthetic flow and climate data, effectively expanding training datasets and enhancing model robustness without the need for additional, resource-intensive simulations [203]. Likewise, domain-specific augmentation methods (employing fluid dynamics similarity laws) can generate many new samples from a single CFD case (by preserving key nondimensional parameters like Reynolds number), thus avoiding numerous costly simulations [204]. Transfer learning and multi-fidelity modeling are also employed: for instance, abundant low-fidelity or related simulation data are used to supplement scarce high-fidelity data, with models fine-tuned in a transfer learning framework [205,206]. Hybrid physics–ML approaches further mitigate data needs by embedding physical knowledge into learning – combining data-driven models with CFD principles to improve generalization from small datasets [207,208].

Another promising strategy is active learning, which iteratively selects the most informative new simulations based on model uncertainty; this has been shown to cut the number of required CFD runs by up to 50% while maintaining accuracy [209]. Finally, domain adaptation techniques seek to bridge the gap between simulated and real-world data: by learning invariant feature representations, models trained on limited synthetic or CFD data can remain robust when deployed under real urban conditions despite distribution shifts [210]. Collectively, these methodologies, such as synthetic data generation, data augmentation, transfer learning, hybrid physics–ML modeling, active learning, and domain adaptation, show substantial potential for alleviating data scarcity constraints, thus enhancing the effectiveness of CFD–ML integration for UGI applications.

The second critical challenge pertains to the limited generalizability and scenario-specific constraints of CFD–ML integration. ML models coupled with CFD often exhibit limited predictive capability beyond their training environments, resulting in poor generalization across different urban morphologies or UGI configurations. Models trained on one particular urban geometry or specific UGI arrangement (e.g., street tree layouts or park designs) may perform inadequately when applied elsewhere unless retrained with additional CFD data representing diverse urban scenarios [211]. Consequently, accuracy can significantly vary with urban form, highlighting the need for continual local

retraining or expansion of training datasets to ensure robust performance in unfamiliar contexts [212].

The third challenge involves potential overfitting to limited or overly simplistic urban descriptors and omission of critical features in CFD–ML modeling for various UGI types. When ML models receive insufficient morphological descriptors, they may inaccurately represent urban permeability and airflow interactions. Wehrle et al. observed overfitting in ML models initially using conventional predictors; substantial performance improvement occurred only after introducing advanced morphometric parameters, such as street orientation and exposure [213]. Thus, integrating CFD outputs with generic ML models without careful, physics-informed feature engineering can be suboptimal. Accurate CFD–ML modeling for diverse UGI scenarios, therefore, requires comprehensive inclusion of critical geometric parameters and underlying physical processes (Tables 1 and 2). For instance, heat mitigation and air quality by UW, ST, GW, and URP are highly context-dependent, strongly influenced by surrounding geometry and ventilation conditions [63,74,84,98]. When essential data is missing or incomplete, direct CFD approaches offer superior physical insight and reliability compared to ML surrogates.

The fourth challenge is that CFD–ML integration faces temporal constraints and limitations in capturing dynamic plant processes. Current CFD–ML studies typically simplify vegetation structures as porous blocks [55,56], and such simplifications inherently limit the precision of ML predictions. If underlying CFD models neglect detailed leaf-level processes (e.g., turbulence and evapotranspiration), subsequent ML predictions become overly idealized and less reliable. Moreover, temporal variations such as seasonal foliage growth or pruning introduce the challenge of concept drift, reducing ML model effectiveness unless periodically retrained with updated data. Hence, temporal changes in vegetation parameters pose significant challenges to the robustness and accuracy of CFD–ML integrated models over extended timeframes.

Finally, it is important to recognize that the heat mitigation and air quality improvement performance of identical UGI varies across different climatic zones [214], potentially influencing the effectiveness of integrated CFD–ML methods. However, this review specifically categorizes integrated CFD–ML applications according to UGI typologies rather than climatic differences. Currently, limited studies have addressed extreme climates, such as polar regions, where vegetation growth is severely restricted by cold temperatures or arid regions, where water scarcity poses significant constraints. Practical implementation of UGI in these climates would necessitate innovative designs, careful species selection, and climate-specific adaptation strategies, explaining the limited research in these contexts. Consequently, UGI research predominantly focuses on temperate or tropical climates, where the benefits such as urban cooling and air purification are extensively documented and comparatively easier to demonstrate.

6.4. Future directions

This review identifies several promising but under-explored future research directions for integrating CFD and ML in UGI studies. Firstly, developing real-time CFD–ML systems that dynamically adjust UGI features (e.g., adaptive shading, misting) based on real-time environmental data could greatly enhance heat mitigation and air quality management. Secondly, unsupervised and semi-supervised learning methods warrant exploration to uncover patterns in large, unlabeled datasets, reducing reliance on costly labeled data and extensive CFD simulations. Techniques such as clustering could identify critical urban morphological and vegetation traits linked to improved cooling and pollutant dispersion. Additionally, hybrid physics-informed ML approaches, transfer learning, and multi-fidelity modeling deserve more attention for robust predictions under limited data and improved generalization across diverse urban scenarios.

Moreover, future studies should explicitly explore CFD–ML integration across different climates, particularly extreme conditions (e.g., arid

or polar regions), to improve model robustness and regional applicability. Finally, establishing standardized guidelines, including optimal dataset sizes, algorithm selection, validation methods, and software compatibility, is essential to facilitate broader, reliable adoption of CFD–ML techniques in urban environmental planning.

Recent advancements in large language models (LLMs) have further expanded the horizon for CFD applications, demonstrating near end-to-end automation capabilities. LLM-driven CFD agents have demonstrated the ability to interpret natural-language specifications, infer boundary and initial conditions, generate consistent OpenFOAM case files, construct computational meshes, initiate solvers, and iteratively correct runtime errors without human intervention [215–217]. For instance, Foam-Agent achieved an executable success rate of 83.6% across a diverse set of 110 benchmark cases [218], while ChatCFD went further, successfully replicating both incompressible and compressible flow scenarios from published studies and autonomously correcting dimensional mismatches, missing files, and numerical scheme errors, maintaining a fully automated success rate of 60–80% for previously unseen cases [217]. Additionally, LLM-based surrogate solvers are emerging as efficient alternatives to traditional CFD methods; notably, the FLUID-LLM fine-tuned a 2.7 billion-parameter OPT model using spatiotemporal adapters, significantly outperforming physics-informed graph neural network baselines by reducing RMSE by 42–48% [219].

Looking forward, these advanced LLM-driven CFD frameworks hold promise for extending into applications involving UGI and urban microclimates, potentially lowering the expertise barrier for non-specialist users. Nevertheless, substantial challenges remain, particularly regarding the handling of complex multiphysics coupling, incomplete boundary-condition reasoning, and limited generalization performance in rare or sparsely documented scenarios. Addressing these challenges will be crucial for maturing LLM-based CFD automation into robust, practical tools for UGI and urban environmental design.

7. Conclusion

This systematic review offers an up-to-date overview of integration strategies between CFD and ML for mitigating urban heat and improving air quality through UGI. It presents the principles, available tools, and coupling approaches for UGI, CFD, and ML. By summarizing, compiling, and discussing recent progress on such integrated strategies, the following conclusions can be drawn:

1. This review examines key UGI parameters—including vegetation type, structure, location, and morphological features—to highlight the crucial factors involved in addressing urban challenges and the transformative role of integrating CFD and ML into UGI design. Researchers and practitioners should carefully consider parameter selection when implementing these integrated strategies.
2. The primary integration methods discussed in the literature are Coupled CFD–ML and Surrogate CFD–ML. In Coupled CFD, ML is embedded within the CFD simulation, making it suitable for real-time augmentation, control, and improvements to turbulence modeling and boundary-condition handling. Surrogate CFD–ML uses CFD runs across scenarios to produce training data; ML models predict new scenarios almost instantaneously. This approach is constructive when exploring extensive design spaces.
3. Many ML algorithms are generally applicable to both air quality and heat mitigation problems. Depending on the four application areas (Fig. 9), researchers should prioritize algorithms that match their data volume and complexity requirements.
4. MLP is the most frequently adopted algorithm due to its adaptability to large datasets or complex relationships. It can handle both classification and regression tasks to predict air pollutant levels or thermal comfort indices, capturing a wide range of urban features.
5. RF and XGBoost are effective for interpretable feature importance without extensive hyperparameter tuning.

6. LR, GPR, and DT are predominantly applied to straightforward air quality predictions.
7. When labelled data are scarce, unsupervised ML techniques may prove beneficial by reducing data complexity.
8. Given the time constraints and the irregularity of UGI vegetation, the RANS turbulence model remains the mainstream approach. Researchers may also consider ROM or dimensionality reduction to decrease degrees of freedom and accelerate simulations while retaining dominant flow characteristics.
9. Where training data are limited, researchers should emphasize data quality and diversity by focusing on key representative scenarios spanning diverse conditions rather than numerous minor variations.

The reviewed literature confirms the significant potential of ML for enhancing CFD simulations of thermal environments and air quality. Although CFD–ML integration demonstrably improves UGI modeling results, most research relies on Surrogate CFD due to the multidisciplinary challenges inherent in Coupled CFD. Overall, there is still limited literature on integrating ML and CFD for UGI, especially regarding heat mitigation and thermal comfort. Future research could explore promising directions such as large language model-driven CFD automation, hybrid physics-informed ML approaches, multi-fidelity modeling, and transfer-learning pipelines, which may further generalise CFD–ML integration for UGI applications. Consequently, the authors recommend combining expertise in fluid mechanics, data science, and urban ecology to select the best ML algorithms and CFD setups for each scenario, thereby maximizing the effectiveness of UGI interventions and advancing our understanding of integration possibilities.

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CRediT authorship contribution statement

Zhu Tao: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hao Sun:** Writing – review & editing. **Bowen Deng:** Writing – review & editing. **Jingxia Wang:** Supervision. **John Calautit:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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